Voiding Dysfunction

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 102	Surgical, Radiographic and Endoscopic Anatomy of the Male Pelvis
Ch 103	Physiology and Pharmacology of the Bladder and Urethra
Ch 104	Pathophysiology and Classification of LUT Dysfunction
Ch 106	Evaluation and Management of Males With Urinary Incontinence
Ch 107	Urodynamic and Videourodynamic Evaluation of the LUT
Ch 109	Neuromuscular Dysfunction of the Lower Urinary Tract
Ch 110	Overactive Bladder
Ch 111	Underactive Detrusor
Ch 113	Pharmacologic Management of Storage and Emptying Failure
Ch 115	Electrical Stimulation and Neuromodulation
★	Putting it together: voiding dysfunction decisions

102 Surgical, Radiographic and Endoscopic Anatomy of the Male Pelvis

1 · Bony pelvis

Why this matters: fixed landmarks for sutures and orientation.

- Sacrum and two innominate bones (**ilium, ischium, pubis** fused at the acetabulum) meet at sacroiliac (SI) joints behind and the symphysis in front.
- **Sacrotuberous** (to ischial tuberosity) and **sacrospinous** (to ischial spine) ligaments resist upward sacral movement; sacrospinous divides the sciatic foramen into **greater and lesser**.
- The pelvic brim is **50–60°** to horizontal. The **arcuate line** (promontory to pubis) separates false from true pelvis. The iliac crest is at **L4**.
- **Cooper's (pectineal) ligament** anchors sutures in hernia repair and urethral suspension.

2 · Lower abdominal wall

Why this matters: fascial fusions direct urine and blood after urethral injury.

Fascia and extravasation

- The umbilicus is **T10** dermatome, at L3. **Scarpa's** (membranous) fascia fuses with fascia lata about 1 cm below the inguinal ligament and continues as **Colles fascia**, then dartos.
- Colles is fused to the ischiopubic rami and perineal membrane, so **anterior urethral injury** fills penis and scrotum with blood and urine: a **"butterfly" perineal haematoma**. They spare the thigh but spread **upward** deep to Scarpa's to the flank.
- †With Buck's fascia intact the haematoma stays on the shaft; a butterfly means Buck's is breached.

Rectus sheath

	Above arcuate line	Below arcuate line
Anterior	External oblique and part of internal oblique	All three aponeuroses
Posterior	Rest of internal oblique and transversus	**Absent**: transversalis fascia and peritoneum only

- Intersections tether rectus to the anterior sheath, so it can be **divided transversely without retracting**.
- Segmental nerves enter **laterally**, so a paramedian incision lateral to rectus causes atrophy and ventral hernia. Pyramidalis is absent in ~20%.

Inguinal canal

- It is **4 cm** long, carrying the **spermatic cord and ilioinguinal nerve**. The deep ring is midway between anterior superior iliac spine (ASIS) and pubic tubercle, **lateral to the inferior epigastric vessels**; the superficial ring is superolateral to the tubercle.
- Walls: external oblique (front), transversalis and conjoint tendon (back), internal oblique and transversus (roof), inguinal and lacunar ligaments and iliopubic tract (floor). The roof closes onto the floor (**shutter mechanism**).
- Direct hernias are **medial**, indirect **lateral** to the inferior epigastrics. At **radical orchidectomy** the cord is ligated at the internal ring, so the stump retracts retroperitoneally.

Peritoneal folds (laparoscopic map)

Fold (medial to lateral)	Contains	Significance
Median	Urachus	Bladder apex to umbilicus
Medial (paired)	Obliterated umbilical arteries	Incise **lateral** to them to reach obturator fossa and retropubic space
Lateral (paired)	Inferior epigastric vessels	Mark deep ring and hernia type

3 · Pelvic floor and fascia

Why this matters: continence support and the planes to the prostate.

- Obturator internus fascia thickens into the **tendinous arch of levator ani** (ischial spine to pubis), origin of the pelvic diaphragm (levator ani, coccygeus).
- Levator ani is **puborectalis** (a U-sling behind the anorectal junction, for **faecal continence**), pubococcygeus and iliococcygeus.
- It is tonically contracted (type I and II fibres), relaxes to void, and **Kegel exercises** increase its tone.
- **Parietal fascia** lines walls; **visceral fascia** sheaths organs; their lateral attachments, the **endopelvic fascia**, carry pelvic vessels and are the gateway to the prostate.
- The **arcus tendineus fascia pelvis** (puboprostatic ligament to ischial spine) marks the parietal–visceral junction and anchors urethral and bladder supports.

4 · Perineum

Why this matters: it carries sphincter nerves and penile vessels.

- The diamond-shaped perineum is divided by the inter-tuberous line into **urogenital** and **anal** triangles. The **perineal membrane** spans the pubic arch, is pierced by the urethra and supports the sphincter.
- The **perineal body** is the midline keystone for pelvic muscles and fascia, blending with Denonvilliers; injury at perineal prostatectomy causes urinary and faecal incontinence.
- The **ischiorectal fossa** opens into the pelvis via the sciatic foramina, so perirectal infection spreads there.
- Perineal prostatectomy MCQ: **Young** divides the central tendon, leaving the external sphincter on the rectum; **Belt** passes **under** it.
- The **internal pudendal artery** exits the greater and re-enters the lesser sciatic foramen, runs in **Alcock's canal**, and becomes the dorsal penile artery. Its veins join the dorsal venous complex (DVC) via the lateral prostatic plexus.
- The **pudendal nerve (S2–S4)** supplies both external sphincters, levator ani, ischio- and bulbospongiosus, ending as the dorsal penile nerve.
- Stretched penile length **approximates erect length**, and is shorter in erectile dysfunction (ED).

5 · Pelvic vessels and lymphatics

Why this matters: pelvic lymph node dissection (PLND) bleeding is predictable.

- Common iliac arteries (in front of the veins) divide at the **SI joint**.

- The **external iliac** gives the inferior epigastric and deep circumflex iliac. **Accessory obturator vessels** from the inferior epigastric cross the obturator fossa (a PLND hazard).
- **Internal iliac**: the posterior trunk is parietal (superior gluteal, ascending lumbar, lateral sacral); the anterior trunk is visceral (umbilical, obturator, middle rectal, inferior vesical, internal pudendal, inferior gluteal).
- The patent umbilical gives the **superior vesical** and artery of the vas (collateral testicular supply). The **inferior vesical** (not internal pudendal) supplies the neurovascular bundle (NVB). The obturator artery is deep and medial to its nerve.
- Iliac veins join **behind the internal iliac artery**: major PLND bleeding.
- The **DVC** enters under the symphysis in front of the striated sphincter; ligating it at prostatectomy can damage the sphincter laterally. **Santorini's plexus** lies under the lateral prostatic fascia.
- Lymph often **crosses sides**. External iliac nodes drain bladder, urachus and wall; internal iliac drain prostate and bladder. The **node of Cloquet** is the **distal limit of PLND**.

6 · Pelvic nerves

Why this matters: each nerve injury has a recognisable deficit.

Nerve	Roots	Key point
Ilioinguinal	L1	Through inguinal canal to anterior scrotum
Genitofemoral	L1–L2	On psoas; **lateral limit of PLND**; genital branch supplies cremaster
Femoral	L2–L4	Injured by psoas hitch, retractor blades, inguinal dissection
Obturator	L2–L4	Iliac bifurcation to obturator canal; PLND injury gives **weak adduction**
Pudendal	S2–S4	Perineal sensation; levator, urinary and anal sphincters
Nervi erigentes	S2–S4	Parasympathetic to bladder, rectum

> **KEY POINT**
> Anterior scrotum is L1 (ilioinguinal, genitofemoral); posterior scrotum is sacral (pudendal, posterior femoral cutaneous).

- P1 Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — *pelvic plexus (T10–L2 + S2–S4) beside rectum at seminal vesicles gives cavernous nerves, NVB outside Denonvilliers at 5 and 7 o'clock*

Not in the paper answer:

- Sympathetic fibres arrive via the superior hypogastric plexus and hypogastric nerves.
- Rectal resection injures the same plexus, so it also causes ED.
- The NVB lies at **5 and 7 o'clock** at the apex; injury at prostatectomy or rectal resection causes ED.

7 · Pelvic viscera

Why this matters: these planes prevent rectal and ureteric injury.

Rectum

- Rectum begins at **S3**. Opening the **rectovesical pouch** exposes the seminal vesicles (posterior robotic approach). **Denonvilliers fascia** separates rectum from prostate.
- Distal rectum near the apex is the **commonest rectal injury site** at prostatectomy.

Bladder

- The bladder is extraperitoneal except the dome, which leads to the **urachus** (adenocarcinoma, fistula, cyst). †The space of Retzius gives extraperitoneal access. The infant bladder is intra-abdominal.
- Detrusor layers are inner longitudinal, middle circular, outer longitudinal. Circular bladder-neck fibres form the **internal sphincter**; these **α1-adrenergic** fibres maintain continence and **antegrade ejaculation**, and damage adds to post-prostatectomy incontinence.
- The body has parasympathetic cholinergic and non-adrenergic non-cholinergic (NANC) fibres *(see Ch 103)*.
- Pedicles are posterolateral, veins drain to the internal iliac, and lymph goes mainly to **external iliac**, also internal iliac and obturator nodes, bilaterally.

Ureterovesical junction

- The intramural ureter runs **1.5–2 cm** obliquely in **Waldeyer's sheath**; being narrow, it traps stones.
- It is a **flap valve**: detrusor backing compresses the ureter. Weak backing causes **vesicoureteral reflux (VUR)**; a weak hiatus allows a **Hutch diverticulum**.
- Superficial trigone anchors at the verumontanum; deep trigone (Waldeyer's and detrusor) at the bladder neck.
- The pelvic ureter narrows at the iliac crossing, under the umbilical artery and **behind the vas**. Blood supply enters **laterally**: keep lateral attachments, as stripping causes stricture.

8 · Endoscopic anatomy

Why this matters: landmarks keep resection safe.

ALGORITHM

1. Under local anaesthetic, inspect bladder, orifices and trigone with a flexible cystoscope.
2. Retroflex at the neck: intravesical prostatic protrusion, hidden tumours.
3. Find the verumontanum and ejaculatory ducts; the external sphincter coapts just distal, then the bulbar urethra.

KEY POINT

The verumontanum is the sentinel at transurethral resection of the prostate (TURP): resection distal to it injures the external sphincter and causes incontinence.

9 · Radiographic anatomy

Why this matters: normal findings mimic disease.

- A **phlebolith** (venous calcification) mimics a distal ureteric or bladder stone.
- **Cystogram**: absent lateral distension suggests extraperitoneal pelvic haematoma or post-prostatectomy lymphocele.
- Voiding cystourethrography (**VCUG**) best shows the posterior urethra, as sphincters relax. On oblique retrograde urethrography (**RUG**), steady pressure overcomes the external sphincter.
- The membranous urethra is the **shortest, narrowest** segment, not a stricture; opacified Littré and Cowper's ducts are normal.
- †Pelvic fracture urethral injury occurs at the bulbomembranous junction.

- Computed tomography (**CT**) has high spatial but low contrast resolution, so needs contrast except for **stones**. **Extraperitoneal rupture** puts urine in retropubic and perivesical spaces.
- Malignant nodes are rounded, heterogeneous; size is most specific at ≥ **1 cm** short axis.
- **CT urogram** excretory images are at **7–10 min**; iliac-crossing narrowing and peristaltic gaps are normal.
- Magnetic resonance imaging (**MRI**) has the best soft-tissue contrast: T1 for bone and gadolinium, T2 for anatomy, prostate, pelvic floor. A distended bladder wall over **3 mm** is thickened. The external sphincter is a T2-dark doughnut below the apex.

103 Physiology and Pharmacology of the Bladder and Urethra

1 · Overview

Why this matters: every voiding drug acts on these mechanisms.

- The lower urinary tract (LUT) has two functions, **storage** (continence) and **emptying** (micturition), integrating the central nervous system (CNS), peripheral nerves, detrusor smooth muscle (DSM), urothelium, urethra, pelvic floor and external urethral sphincter (EUS).

> **KEY POINT**
> **Why single-organ therapy fails:** bladder and urethra act reciprocally, so boosting contractility in detrusor underactivity fails unless the urethra also relaxes. Afferent modulation acts on both (reduce it for overactivity, augment it for underactivity), which makes it the most promising therapeutic lever.

2 · Functional anatomy

Why this matters: structure explains storage, sphincter strength and stiff bladders.

Bladder

- The **body** lies above the ureteric orifices; the **base** (trigone and bladder neck) differs in neuropharmacology.
- The **male bladder neck** is a complete smooth muscle ring with dense sympathetic supply; it closes in ejaculation, ensuring **antegrade ejaculation**. Women lack this collar and have sparse adrenergic supply.
- The **lamina propria** is the "functional centre", with the afferent plexus (TRPV1, P2X3), interstitial myofibroblasts and subepithelial capillaries.
- **Blood flow** falls with distension (wall tension), more so in low compliance. Ischaemia–reperfusion causes patchy denervation, as in detrusor overactivity (DO).

Stroma

- The wall is about **50% collagen, 2% elastin**, mainly types I, III and IV.
- **Poorly compliant bladders** have more connective tissue relative to muscle and a higher **type III : type I collagen** ratio, so capacity falls.

Urethra and sphincters

- The urethra has smooth muscle (lissosphincter) and an **intramural striated rhabdosphincter (EUS)**, separate from the periurethral pelvic floor.
- The male EUS is a crescent in front of the prostate, becoming a horseshoe distally. The female EUS is a horseshoe over the ventral urethra, and a subepithelial vascular plexus forms a **"urethral seal"**.
- Levator ani does not encircle the ventral urethra, so its direct role in continence is questioned.

Feature	Slow-twitch	Fast-twitch
Role	Sustained resting tone	Rapid extra squeeze on coughing

Feature	Slow-twitch	Fast-twitch
Fatigue / metabolism	Slow; oxidative	Rapid; anaerobic
Where	Intramural rhabdosphincter — **~50% of static urethral resistance**	Periurethral pelvic floor (mixed)
Rhabdosphincter mix	Male **65% slow / 35% fast**; female **87% slow / 13% fast**	

3 · Urothelium: barrier and sensor

Why this matters: the urothelium senses stretch; its damage drives urgency and pain.

Barrier

- About 7 layers are topped by **umbrella cells**, whose **uroplakins** form **asymmetrical unit membrane** plaques: the barrier, and the binding site of **type 1-fimbriated *E. coli***.
- **Tight junctions** (occludin, claudins, ZO-1) block paracellular flow; stretch lowers ionic resistance tenfold without changing small-molecule permeability.
- Loss of umbrella cells (infective or radiation cystitis) lets solutes leak in, so symptoms worsen with distension; hyperosmolality excites C fibres.
- In the **glycosaminoglycan (GAG) layer**, the "antiadherence factor" is mucin MUC-1 and chondroitin sulfate is the main GAG; it is **not proven** as the primary barrier.

Sensor–transducer function

Urothelial cells behave like sensory neurons, with receptors for bradykinin, TrkA/p75, P2X/P2Y, α/β, nicotinic/muscarinic, ENaC and TRPV1/2/4, TRPM8. On stretch they release mediators onto afferents, directly or via myofibroblasts.

Mediator	Released by / when	Effect
ATP	Stretch; more with injury, inflammation, ageing	Excites suburothelial **P2X3** afferents, giving fullness and pain; P2X3-null mice are hyporeflexic
Nitric oxide (NO)	Especially inflammation	Blocks afferent Ca^{2+} channels: **afferent inhibition**
Acetylcholine (ACh), non-neuronal	Basal; more with stretch, ageing	Modulates afferents and DSM, so antimuscarinics may act partly here
Prostaglandins (PGs)	Urothelium is the main source	Detrusor regulation, cytoprotection
Inhibitory factor (unknown)	Basal	Reduces muscarinic contraction; not NO or PG

- The afferent threshold is set by the **ATP (excitatory) : NO (inhibitory) balance**. ATP is raised in DO and in bladder pain syndrome/interstitial cystitis (BPS/IC).
- **Suburothelial myofibroblasts** (vimentin+, α-smooth muscle actin+, desmin−) are linked by connexin 43 (**Cx43**) gap junctions and respond to ATP (P2Y6) and ACh (M2/M3); M2/M3 expression correlates with urgency in idiopathic DO.

> **KEY POINT**
> Damaged urothelium causes symptoms twice over: by **solute leak** into the lamina propria and by **altered signalling** (more ATP, less NO) to afferents.

4 · Detrusor smooth muscle physiology

Why this matters: each contraction step is a drug target.

- Fascicles run randomly without sarcomeres; actin and myosin anchor to **dense bodies**. Not every cell is innervated, so **gap junctions** spread excitation.
- Unlike skeletal muscle, control is autonomic and paracrine, and Ca^{2+}–calmodulin drives **myosin phosphorylation** (slow, "latch state"); activity is mostly tonic.

ALGORITHM

1. ACh binds **M3** receptors (Gq).
2. Intracellular Ca^{2+} rises via L-type entry and phospholipase C (PLC), whose IP3 releases sarcoplasmic reticulum (SR) Ca^{2+}.
3. Ca^{2+}–calmodulin activates myosin light chain kinase (MLCK), which phosphorylates MLC20.
4. Cross-bridges cycle and the muscle contracts.
5. As Ca^{2+} falls, myosin light chain phosphatase (MLCP) dephosphorylates MLC20 and the muscle relaxes.

- Tone depends on Ca^{2+} and the **MLCK : MLCP balance**. In **calcium sensitisation**, RhoA-activated **Rho kinase** and protein kinase C (PKC, via CPI-17) inhibit MLCP, so force rises without more Ca^{2+}.
- **L-type voltage-dependent Ca^{2+} channels (VDCC)** are the main trigger for contraction (blocked by nifedipine, diltiazem, verapamil); T-type channels give fine control.
- Resting potential is −50 to −60 mV; Ca^{2+} carries the upstroke and K^+ (SK, **BK**, Kv) repolarises. **BK is the key brake**: Ca^{2+} sparks open BK, hyperpolarising the cell. BK knockout gives overactive bladder (OAB); BK is reduced in neurogenic DO (NDO).

G protein	Receptors	Effect
Gq	M1, M3, M5 ("odd"); α1; AT II; H1; 5-HT2	IP3 and DAG release Ca^{2+} and activate PKC: contraction
Gi/o	M2, M4 ("even")	Lower cAMP
Gs	β3	Raise cAMP, activating PKA: relaxation

- **Purinergic (P2X1) contraction** is minimal normally but appears in obstructed, OAB and ageing bladders. It suppresses muscarinic coupling, explaining detrusor hyperactivity with impaired contractility (**DHIC**): urgency with incomplete emptying.
- The detrusor is deliberately **poorly coupled** (no tetanic contractions), protecting filling. More Cx43 (bladder outlet obstruction [BOO], NDO; induced by cytokines and nerve growth factor [NGF]) allows whole-bladder myogenic contraction, which is DO.
- **Interstitial cells** are c-KIT+ (imatinib reduces spontaneous contractions; increased in OAB, BOO) or inhibitory PDGFRα+ (P2Y1).

KEY POINT

Motor–sensory coupling: spontaneous micromotions generate afferent bursts perceived as urgency. Antimuscarinics and mirabegron reduce non-voiding contractions and so reduce sensory input; efferent and afferent functions are inseparable.

5 · Receptors and drug targets

Why this matters: the basis of every bladder drug.

Receptor	Location	Function	Drug target
M3 (Gq)	Detrusor; urothelium; salivary glands	**Mediates contraction**	Antimuscarinics
M2 (Gi)	Detrusor (**numerically predominant**)	Amplifies M3 by opposing β-cAMP relaxation; the balance shifts from M3 to M2 in neurogenic and obstructed bladders	Non-selective antimuscarinics
β3 (Gs)	Detrusor (**most expressed adrenoceptor**); afferents	Relaxes during storage; increases compliance	Mirabegron, †vibegron
α1A	Urethra, prostate	Urethral closure; female resting tone	α-blockers (benign prostatic hyperplasia, BPH); α-agonists increase closure
α1D	Bladder (predominant α); spinal cord	Storage symptoms of obstruction	Tamsulosin improves storage lower urinary tract symptoms (LUTS)
α2	Prejunctional urethra; ganglia	Reduces noradrenaline release	—
P2X3	Suburothelial afferents	Senses ATP; increased in BPS/IC	AF-219 (proof-of-concept)
P2X1	Detrusor	Purinergic contraction; increased in BOO, ageing	Potential for DHIC
TRPV1	C fibres; urothelium	Senses heat >43 °C, protons, capsaicin; increased in NDO	Capsaicin and resiniferatoxin (RTX) desensitise
TRPM8	Afferents	Senses cold (bladder cooling reflex)	Experimental
NK1	Spinal cord	Afferent transmission	Aprepitant reduced urgency (trial)
Nicotinic	EUS	Sphincter contraction	—

- P3 OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron — *tissue > subtype selectivity; darifenacin ~30-fold M3; β3 raises cAMP; dementia and antimuscarinic burden*

Not in the paper answer:

- M2 dominance grows in neurogenic and obstructed bladders.
- Urothelial muscarinic receptors mean antimuscarinics partly damp afferents.
- **Oxybutynin**: its metabolite **desethyloxybutynin** causes most of the dry mouth, so the controlled-release form is better tolerated.
- CNS-penetrant agents block M1, causing delirium and cognitive decline, reversible within weeks of stopping.
- Other storage drugs: Ca^{2+} antagonists, K^{+} openers, imipramine (anticholinergic, direct relaxant, noradrenaline reuptake blocker), capsaicin and RTX.
- Outlet drugs are **α-agonists** (ephedrine, midodrine). **Oestrogen** improves seal and blood flow but disappoints alone; systemic oestrogen may worsen urinary incontinence (UI).

- P3 Duloxetin — *SNRI raises Onuf's nucleus output and so sphincter tone; mild–moderate SUI; Europe, not USA*

6 · Peripheral innervation

Why this matters: nerve roles predict the bladder after injury.

- P1 Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children *— pelvic (S2–S4) voiding, hypogastric (T10–L2) storage, pudendal/Onuf's EUS; Aδ vs silent C fibres; storage spinal vs spinobulbospinal voiding reflex; PAG–PMC; central transmitters*

Not in the paper answer:

- Pelvic and pudendal afferents enter sacral, hypogastric afferents lumbar, dorsal root ganglia.
- The PMC relaxes the sphincter only indirectly; the L-region stores (section 7). Spinal shock precedes the C-fibre reflex (section 8).
- Parasympathetic outflow is **S2–S4**; sympathetic is †**T10–L2**, relaying in chain, inferior mesenteric and pelvic ganglia.
- Some parasympathetic postganglionic neurons are **intramural**, so cauda equina or pelvic plexus injury **decentralises rather than denervates** the bladder.
- The afferent plexus is densest at the bladder neck, absent at the dome. Filling gives episodic bursts; the last is felt as urgency.
- Aδ fibres are capsaicin-resistant muscle tension receptors. C fibres are mucosal, mostly **silent**, capsaicin/RTX-sensitive, and drive a new **spinal reflex** after spinal cord injury (SCI), inflammation, BOO or ageing.
- **Sensitisation** by ATP, NGF (TrkA), PGs, bradykinin, low pH and high osmolality leads on to central sensitisation. **Cross-organ sensitisation** explains BPS/IC overlapping irritable bowel syndrome.
- μ-opioids reduce sensation and raise sphincter tone, causing postoperative retention.

7 · Central control

Why this matters: lesion level predicts the bladder pattern.

- The pontine micturition centre (**PMC**) is **Barrington's nucleus** ("M region", dorsal pontine tegmentum); lesions (stroke, multiple sclerosis, tumour) cause **retention**. It does **not** project to Onuf's nucleus but relaxes the sphincter **indirectly** via GABA/glycine interneurons.
- The **L-region** (ventrolateral pons) is the storage centre and projects to Onuf's nucleus.
- The periaqueductal grey (**PAG**) relays afferents to the PMC, forming a binary "on–off" switch; without higher control the bladder empties at a critical volume (infants).
- **Forebrain**: signals pass from PAG via thalamus to the **insula** (desire) and **prefrontal cortex** (decision), which keeps the reflex under **tonic suppression**.
- Frontal white-matter lesions cause **incontinence**. In urge incontinence the anterior cingulate and supplementary motor area (SMA) over-respond.

> **KEY POINT**
> **Lesion localisation:** a forebrain lesion removes tonic inhibition, giving DO with preserved coordination. A pontine lesion gives retention. A suprasacral cord lesion gives DO with detrusor–sphincter dyssynergia (**DSD**). A sacral or cauda lesion gives a decentralised, acontractile bladder *(see Ch 104)*.

8 · Storage and emptying phases

Why this matters: these reflexes explain DSD and neuromodulation.

- Storage pressure stays low through intrinsic wall properties, quiet parasympathetic outflow and a **sympathetic storage reflex**, which is weak in humans (bilateral retroperitoneal lymph node dissection does not alter storage).
- **Guarding reflex**: filling recruits pudendal motoneurons, so EUS electromyography (EMG) activity rises. On coughing, urethral pressure rises **before** pressure transmission, proving an active neural component. Its suppression in voiding is supraspinal, so SCI causes **DSD**.
- **Sphincter-to-bladder reflex**: pudendal afferents inhibit bladder preganglionic neurons, the basis of "holding on" and neuromodulation.

ALGORITHM

1. Tension-receptor (Aδ) firing exceeds threshold.
2. If the cortex permits, the PAG/PMC switch flips.
3. Sympathetic and pudendal outflow switch off; the **urethral sphincter relaxes first**, and urethral smooth muscle relaxes (NO).
4. The detrusor contracts (parasympathetic) and flow begins.
5. Urethral afferents reinforce the contraction until the bladder is empty.

- **Urethra-to-bladder reflexes** (Barrington): urethral flow facilitates contraction, linking stress urinary incontinence (SUI; urine in the proximal urethra) with detrusor instability.
- After SCI, weeks of spinal shock (retention) precede an NGF-driven **C-fibre spinal reflex**, giving NDO with DSD.
- **Pdet = Pves − Pabd**. Detrusor power (Pdet × flow, Q) is roughly constant, so voiding **Pdet reflects outlet resistance, not contractility**.
- Women with SUI void at Pdet at Qmax ≈ **19 cmH₂O**, healthy men ≈ **60–70 cmH₂O**. Low pressure in women does not mean impaired contractility; male nomograms are invalid in women.

9 · Urethral closure mechanisms

Why this matters: each SUI treatment targets one closure force.

- Nicotinic blockade rarely lowers urethral tone by >40%, but α-blockade lowers urethral pressure by about **one third**, so smooth muscle (α1A) tone matters.
- The EUS closes uniformly like an **iris**, and noradrenaline augments it.
- **Mucosal seal**: lamina propria vessels fill to seal the lumen; oestrogen increases flow, and poor arterial supply lowers urethral pressure.
- **Passive pressure transmission** reaches the proximal urethra; in the **hammock hypothesis (DeLancey)** it is compressed against a firm posterior fascial layer.
- **Active** guarding adds a fast-twitch reflex squeeze on coughing.

KEY POINT

Continence = **active muscle tone + passive anatomical coaptation**. The integral theory led to the mid-urethral sling (MUS), the SUI standard, yet DeLancey's data suggest intrinsic sphincter function matters more than support *(see Ch 109)*.

10 · Compliance

Why this matters: poor compliance damages kidneys.

- P3 Urinary bladder compliance — *$C = \Delta V/\Delta P$; collagen/elastin, coiled collagen straightens; fast filling falsely low; DLPP >40 cmH_2O means upper tract damage*

Not in the paper answer:

- Poorly compliant bladders cannot uncoil their collagen.
- Normal filling keeps pressure **<10 cmH_2O**. Determinants are viscoelasticity, active tone and geometry; **type III collagen straightens** with filling.
- Pelvic nerve (but not acute spinal) transection reduces compliance, showing that an afferent–CNS loop inhibits contractility during filling.
- †A detrusor leak point pressure (DLPP) ≥ **40 cmH_2O** in a low-compliance neurogenic bladder predicts upper tract damage *(see Ch 104)*.

11 · Pathophysiology links

Why this matters: why symptoms persist and drugs fail.

- **Idiopathic OAB**: DO is found in a minority (11% in one cohort), so symptoms reflect dysregulated sensation ("afferent neurourology"): more ATP/TRPV1 and Cx43, less BK, NGF plasticity.
- **BOO** causes hypertrophy and partial denervation, which lead to **muscarinic supersensitivity** and more P2X1 and Cx43 (myogenic theory, Brading). NGF stays raised, which explains **persistent storage symptoms after prostate surgery**.
- **BPS/IC** shows more NGF, ATP/P2X3, TRPV1 and mast cells; tanezumab and intravesical RTX failed (heterogeneity) *(see Ch 113)*.
- **Underactive bladder (UAB)** is a symptom complex; detrusor underactivity is the urodynamic finding, and afferent loss may contribute.
- A post-void residual (PVR) ≥ **300 mL** warrants evaluation, as do bothersome LUTS, hydronephrosis, renal insufficiency or culture-proven urinary tract infection (UTI); PVR <300 mL does not prove normal contractility.

12 · Ageing

Why this matters: the aged bladder is weak yet irritable.

- Ageing causes **hypofunction**: contractility, sensation and closure pressure fall. Capacity does not fall per se; small capacity means coexisting DO.
- Muscle falls relative to collagen, with ischaemia and cholinergic nerve loss. **α-contraction increases (α1D) and β-relaxation decreases**, so noradrenaline contracts the aged bladder.
- The aged CNS is prone to overactivity, giving urgency with poor emptying (DHIC). †Prefer β3 agonists or low-CNS-penetrant agents in the elderly.

13 · Neuromodulation and botulinum toxin — mechanisms

Why this matters: one stimulator treats overactivity and retention.

- Sacral neuromodulation (**SNM**), pudendal stimulation and posterior tibial nerve stimulation (**PTNS**) treat refractory OAB **and** non-obstructive retention. Their target is **afferent axons**.

- In OAB, somatic input inhibits preganglionics and transfer to the PMC; in retention, it switches off an overactive **guarding reflex**.
- Pudendal stimulation carries S2–S4 afferents, avoiding sciatic discomfort from S3.
- **OnabotulinumtoxinA (OBTX)**: the heavy chain binds the terminal, and the light chain (zinc protease) cleaves **SNAP-25**, reducing ACh release. It also lowers TRPV1/P2X3, substance P, CGRP and urothelial ATP.
- RTX desensitises TRPV1 **without prior excitation**, unlike capsaicin.

14 · Landmark trials

Why this matters: toxin versus tablets and SNM.

Trial	Design	Result	Take-home
ABC	RCT: intradetrusor OBTX vs oral antimuscarinic, idiopathic OAB with urgency urinary incontinence (UUI)	Equal fall in daily UUI; OBTX gave complete dryness more often	Antimuscarinic: dry mouth; OBTX: catheterisation, UTI
ROSETTA	RCT: OBTX 200 U vs SNM, women with refractory non-neurogenic OAB, 2 years	Comparable efficacy; OBTX more often dry	OBTX more UTIs (note 200 U dose)

104 Pathophysiology and Classification of Lower Urinary Tract Dysfunction

1 · Normal function: two phases

Why this matters: every dysfunction is a failure of one of these requirements.

Storage requires	Voiding requires
Low detrusor pressure (normal compliance), appropriate sensation	Detrusor contraction of adequate magnitude and duration
Outlet closed at rest and on straining	Fall in smooth and striated sphincter resistance
No detrusor overactivity (DO)	No anatomic obstruction

- **Storage is mostly passive** (viscoelastic stroma). More collagen (inflammation, bladder outlet obstruction [BOO], decentralisation) gives **drug-unresponsive** low compliance.
- Storage reflexes: α1 bladder-neck tone, β3 detrusor relaxation, spinal **guarding reflex**. Voiding uses pelvic parasympathetic outflow, coordinated by the **pons**.
- **Stress continence** needs equal pressure transmission to the proximal urethra (**Enhorning**), active reflex closure (**Tanagho**) and a **mucosal seal** (poor if scarred or atrophic). **Klevmark**: cystometric pressure rise is a fast-filling artefact.

2 · Pathophysiology

Why this matters: naming the failing part leads straight to treatment.

Every failure is of the **bladder, outlet, or both**: the basis of the **Wein functional classification**.

Storage failure

- **Bladder overactivity** means involuntary contractions, low compliance, or both (neurological, BOO, ageing, idiopathic).
- **Overactive bladder (OAB)** (International Continence Society, ICS) is urgency with or without urgency urinary incontinence (UUI), usually with frequency and nocturia. With UUI assume DO, as urodynamic studies (UDS) often miss it.
- **Hypersensitivity** without overactivity occurs in interstitial cystitis/bladder pain syndrome (IC/BPS). **Outlet underactivity** follows denervation, surgery, trauma, ageing or loss of female support.

	Hypermobility (genuine stress urinary incontinence, SUI)	Intrinsic sphincter deficiency (ISD)
At rest	Outlet competent	Bladder neck poorly functional or open
Cause	Poor pelvic support	Prior surgery, scarring, neurological
Treatment	Correct hypermobility	Coaptation or compression

KEY POINT

Urinary incontinence (UI) can occur from ISD alone, but **never from hypermobility alone**. In **DeLancey's hammock**, a firm suburethral layer lets abdominal pressure compress the urethra. In half of continent women the continence point is the **mid-urethra**. Male sphincteric incontinence resembles ISD.

Emptying failure

- **Bladder underactivity** is neurogenic, myogenic (overdistension, drugs) or psychogenic; diabetes causes neuromuscular and vascular damage.
- **BOO** (ICS) is low flow and/or raised post-void residual (PVR) **with high detrusor pressure**. Male causes: prostate, bladder neck contracture, stricture; female: prolapse, post-sling.
- **Functional** obstruction: striated sphincter dyssynergia (commonest), smooth sphincter dyssynergia (**autonomic dysreflexia**), primary bladder neck obstruction (young men) and dysfunctional voiding (**non-neurogenic**).

3 · Functional classification (Wein) — 2 × 2 table

Why this matters: the table maps each failure to its treatment family.

	Because of the bladder	Because of the outlet
Failure to store	DO; low compliance; hypersensitivity; underactivity with overflow	Hypermobility; ISD. (Plus **fistula**)
Treatment	Bladder training; antimuscarinics, β3-agonists, onabotulinumtoxinA; neuromodulation; augmentation	Pelvic floor training; α-agonists, duloxetine; bulking; slings; artificial urinary sphincter (AUS)
Failure to empty	Underactivity: neurogenic, myogenic, psychogenic	Anatomic obstruction; sphincter dyssynergia; bladder neck dysfunction
Treatment	Valsalva/Credé; parasympathomimetics; electrical stimulation	α-blockers, 5α-reductase inhibitors (5ARIs); prostatectomy, incision; sphincter onabotulinumtoxinA, sphincterotomy
Circumvent	Pads or collecting devices, antidiuretics, catheterisation, diversion	

KEY POINT

- Bladder underactivity appears under **both** storage (overflow) and emptying. MCQ
- An α-agonist tightens the outlet (storage); an α-blocker relaxes it (emptying).
- OnabotulinumtoxinA in the detrusor treats storage; in the sphincter it treats emptying.
- Naming the failure names the treatment family.

Combined deficits (T10 paraplegia: DO and dyssynergia) are managed by external collection, or by **abolishing DO** plus clean intermittent catheterisation (CIC).

4 · Urodynamics by phase and site

Why this matters: choose the test that answers the question.

Phase	Bladder	Outlet
Storage	Filling cystometry; detrusor leak point pressure (DLPP)	Urethral pressure profile; Valsalva leak point pressure (VLPP)
Voiding	Voiding cystometry (Pdet)	Fluoroscopy; sphincter electromyography (EMG)

Phase	Bladder	Outlet
Both	Flowmetry and PVR **cannot localise** the fault	

5 · Other classifications

Why this matters: ICS terms are the language of exams and reports.

- P2 Pathophysiology, classification and management of LUTS. — *ICS storage / voiding (incl. terminal dribble) / post-micturition (post-micturition dribble) symptoms; Wein summary*

Not in the paper answer:

- Flow and PVR cannot separate a weak detrusor from obstruction.
- Chronic obstruction causes hypertrophy, collagen, diverticula and DO, so intervene promptly.
- **Signs (ICS)**: bladder diary, examination (prolapse may mask UI), pad test.

ICS urodynamic classification (most precise)

Domain	Storage	Voiding
Detrusor	Normal; DO (neurogenic or idiopathic)	Normal; underactive; acontractile
Sensation, capacity, compliance	Normal, high/increased, low/reduced, absent	—
Urethra	Normal or incompetent closure	Normal; BOO, dysfunctional voiding, detrusor–sphincter dyssynergia (DSD)

- Cystometric capacity is when permission to void is given; **maximum cystometric capacity is when one can no longer delay**; maximum anaesthetic capacity is under anaesthesia.
- DSD is detrusor contraction with involuntary striated sphincter contraction; add "neurogenic" when a relevant neurological disorder exists.

† **Lapides**: sensory neurogenic, motor paralytic, uninhibited, reflex, autonomous. **Bors–Comarr**: upper/lower motor neuron, complete/incomplete, balanced/unbalanced. MCQ

6 · Nocturia and neurogenic dysfunction

Why this matters: lesion level predicts the bladder and sphincter pattern.

- **Polyuria** is >**40 mL/kg/24 h**. **Nocturnal polyuria** is a night-to-24-h volume ratio >**20% young, >33% elderly**; causes are loss of the nocturnal vasopressin rise and sleep apnoea.
- In **IC/BPS**, normally silent **C fibres** become hyperactive (central sensitisation), while Aδ fibres signal distension. Bladder–bowel crosstalk explains OAB–irritable bowel overlap.

Lesion	Detrusor	Sphincter
Suprapontine	DO	Synergic
Suprasacral spinal	DO	DSD, raised PVR
Sacral	Acontractile	Impaired

> **KEY POINT**
> DSD needs an intact sacral arc **disconnected from the pons**, so it is suprasacral, never suprapontine.

106 Evaluation and Management of Males With Urinary Incontinence

1 · Definition and determinants

Why this matters: find the cause before treating.

- **Urinary incontinence (UI)** (International Continence Society, ICS) is any involuntary loss of urine, a storage lower urinary tract symptom (LUTS) *(see Ch 104)*. In men the **external urethral sphincter (EUS) is most important**.
- Risk factors: **age (major, independent)**, LUTS, infection, cognitive or neurological impairment, obesity, diabetes, prostatectomy, drugs, smoking.
- Reversible causes, **DIAPPERS**: Delirium, Infection, Atrophic urethritis, Pharmacological, Psychiatric, Endocrine, Reduced mobility, Stool impaction.
- Male:female ≈ 1:2. **Urgency UI (UUI) predominates**, then stress UI (SUI), then mixed UI (MUI). Causes are often multiple, so establish aetiology first.

2 · Types

Why this matters: the leak type points to treatment.

Type (ICS)	Mechanism	Male context
SUI: on effort, cough	Sphincter deficiency	Almost only **after prostatectomy**
UUI: with urgency	Detrusor overactivity (DO) overcomes a normal outlet	~**90%** have DO on urodynamics; it is "wet" overactive bladder (OAB)
MUI	DO and sphincter deficiency	Uncommon; may follow prostatectomy
Nocturnal enuresis	Wetting in sleep	New in an adult man: suspect **high-pressure chronic retention** and renal failure
Continuous	Fistula, exstrophy/epispadias, ectopic ureter	Rare
With chronic retention	Residual ≥**300 mL**	"Overflow" is no longer an ICS term; spinal cord injury (SCI) at or above T6 risks autonomic dysreflexia

- Others: postural, insensible, **postmicturition dribble**, functional, situational.

3 · Evaluation

Why this matters: most men need only simple tests.

- History: timing, triggers, containment, prior prostate or pelvic surgery, neurology, bowel and sexual function, drugs, **most bothersome symptom**.
- Examine abdomen, genitalia (meatal stenosis, phimosis), digital rectal examination (DRE), anal tone, neurology and cognition; do a **cough test supine then standing**.
- **Body mass index (BMI)**: weight loss may help SUI. With an artificial urinary sphincter (AUS), abdominal pressure above balloon pressure (61–70 cm H_2O) leaks through the cuff.
- **Dipstick** all men; haematuria needs cytology, cystoscopy and upper-tract imaging.

- **Bladder diary**: only the ICIQ Bladder Diary (ICIQ-BD) is fully validated. **3 days equal 7 days**; include a weekend.
- The International Prostate Symptom Score (IPSS) does **not measure incontinence**; use the **ICIQ-UI Short Form** (0–21). A **pad test** is not routine and cannot separate UUI from SUI.

Severity	Pads/day	Pad weight/day
Mild	1	<200 g
Moderate	2–4	200–400 g
Severe	>4	>400 g

- Pad weight beats pad count; neither is validated.
- **Flow and post-void residual (PVR)**: a fast rise suggests OAB; a slow rise with a long tail suggests bladder outflow obstruction (BOO). Flow cannot separate BOO from underactivity. One representative flow is acceptable (ICS 2017).
- Check prostate-specific antigen (PSA) only if life expectancy is >10 years and it would change management.
- Imaging or cystoscopy: high PVR, haematuria, stricture. Do urodynamic studies (**UDS) only if they alter management**, i.e. before invasive treatment (International Consultation on Incontinence [ICI], EAU, AUA guidance) *(see Ch 107)*.
- SUI is a symptom; **urodynamic stress incontinence (USI)** is leakage on effort without detrusor contraction. **Cough-associated DO incontinence** mimics SUI, so fill supine and cough: leak without DO is USI.

> **KEY POINT**
> Every man: history, examination, BMI, dipstick, diary, questionnaire, flow, PVR. Optional: PSA, bloods, imaging, endoscopy, invasive UDS, usually just before surgery.

4 · Urgency urinary incontinence

Why this matters: the commonest male type, treated stepwise.

> **ALGORITHM**
> 1. Lifestyle, bladder training and pelvic floor muscle training (PFMT) ≥6 weeks (ideally 3 months).
> 2. Antimuscarinic or β3-agonist, or both.
> 3. Urodynamics optional; then onabotulinumtoxinA, sacral neuromodulation (SNM) or posterior tibial nerve stimulation (PTNS).
> 4. Last resort: augmentation or ileal conduit.

- Cut fluid by **25%** if intake is >1 L/day; avoid caffeine and fizzy drinks; lose weight.
- **Antimuscarinics** have similar efficacy; extended-release is preferred. **Anticholinergic burden** ≥3 months raises dementia risk ~50% and causes falls. Trospium has little brain penetration *(see Ch 113)*.
- **β3-agonists** are co-first-line with less dry mouth. Mirabegron: **monitor blood pressure**; vibegron: no CYP450 interaction. **Cognitive impairment: β3-agonist or early surgery**.
- Mirabegron plus solifenacin beats escalating solifenacin. An α-blocker plus antimuscarinic or β3-agonist suits mixed LUTS. Tadalafil is the only phosphodiesterase-5 inhibitor for LUTS with erectile dysfunction.

	OnabotulinumtoxinA	SNM	PTNS
Delivery	Cystoscopic, **100 U** (200 U neurogenic DO)	S3 lead, implanted generator	Needle behind medial malleolus, 30 min weekly × 12
Strength	Repeat every 6–9 months	Most effective, durable >10 years	Least invasive; better for dry OAB
Drawback	Retention in ~**10%**, so the man must accept self-catheterisation	Most invasive, costliest	Frequent visits

- If the SNM test gives ≥**50%** improvement, a permanent generator is implanted *(see Ch 115)*. No head-to-head trials exist.
- Refractory cases need containment, or augmentation or ileal conduit (high morbidity).

5 · Stress urinary incontinence

Why this matters: surgery is matched to severity.

ALGORITHM

1. Supervised PFMT ≥3 months, with or without biofeedback and electrical stimulation (better started pre-op); consider duloxetine.
2. Exclude anastomotic or urethral stricture and bladder dysfunction (UDS).
3. Choose surgery by severity.

- P2 Post-prostatectomy incontinence — *wait 6–12 months + PFMT; duloxetine off-label; no bulking; sling mild–moderate, AUS (cuff, balloon, scrotal pump) severe/irradiated; ProACT; MASTER*

Not in the paper answer:

- Prior transurethral surgery worsens fixed-sling results; ProACT is explanted in up to half.
- EAU 2023 grades: AUS strong, transobturator sling weak (table below).
- After radical prostatectomy wait **6–12 months**; avoid a long-term urethral catheter (traumatic hypospadias).
- The **AUS is the gold standard**, ~80% continence (Scott 1973 MCQ), activated at ~6 weeks; †deactivate it before catheterisation or endoscopy to prevent erosion.
- Slings are fixed transobturator (AdVance) or adjustable (Remeex, Argus-T, ATOMS); neither is superior. Radiotherapy, severe leakage and prior transurethral resection worsen fixed-sling results, so choose an AUS.
- **ProACT** puts balloons at the bladder neck; up to half are explanted; **contraindicated after radiotherapy**.
- Neurogenic SUI often needs several procedures *(see Ch 109)*.

EAU 2023 recommendation (post-prostatectomy incontinence, PPI)	Grade
Offer AUS for moderate–severe PPI	Strong
Warn of AUS/ProACT complications, mechanical failure, explantation	Strong
Inform of duloxetine side effects, off-label use	Strong
Fixed transobturator sling for mild–moderate PPI	Weak
Do not offer bulking agents; no ProACT after radiotherapy	Weak

6 · Mixed and other types

Why this matters: each pattern has one key rule.

- **MUI**: treat the most bothersome component first; DO may mask SUI, so treat DO and repeat UDS.
- High-pressure chronic retention with enuresis: **catheterise first**, then relieve BOO.
- Enuresis with low residual is usually OAB: antimuscarinic or β3-agonist, with or without desmopressin.
- Postmicturition dribble: pelvic floor squeeze after voiding and **urethral milking** (press behind the scrotum forward to the penile base).
- **Red flags** need a specialist: haematuria, pain, recurrent infection, pelvic radiotherapy, abnormal DRE, high PVR. AUS: ICI grade A.

7 · Landmark trials

Why this matters: a sling can replace an AUS.

Trial	Design	Result	Take-home
MASTER	Non-inferiority randomised controlled trial (RCT), male sling vs AUS for urodynamic SUI after prostatectomy	Equivalent at 12 months; secondary outcomes favoured AUS; few completely dry	Sling is an acceptable alternative; counsel realistically

107 Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract

1 · Definition and role

Urodynamics answers a set question about the lower urinary tract (LUT).

- **Urodynamics (UDS)** is the dynamic study of transport, storage and evacuation of urine. The International Continence Society (ICS) separates UDS *studies* (tests) from *observations* (findings).
- Use it when history, examination and simple tests fail: (1) to **diagnose the cause of symptoms**; (2) to assess **diseases that silently damage the upper tracts** — spinal cord injury (SCI), multiple sclerosis (MS), radiation cystitis. It also predicts outcome and explains treatment failure.

Wein functional classification — the frame for every study

Site	Overactive	Underactive
Bladder	Failure to store	Failure to empty
Outlet	Failure to empty	Failure to store

- Set the questions first, then design the study, aiming to **reproduce the symptom**. An abnormality without symptom reproduction is not necessarily diagnostic; failure to record one does not exclude it.

KEY POINT

Box 107.1 — Urodynamic risk factors (treat even if asymptomatic): impaired compliance; detrusor–external sphincter dyssynergia (DESD); detrusor–internal sphincter dyssynergia (DISD); high-pressure detrusor overactivity (DO) throughout filling; detrusor leak point pressure (DLPP) **> 40 cmH_2O**; poor emptying with high storage pressure. Each causes sustained high intravesical pressure.

2 · Preparation

Drugs, baseline flow and infection decide whether the test is valid and safe.

- **Decide medication status in advance**: test on regular dosing to assess drug effect; stop the drug first to uncover aetiology.
- **ICS pre-invasive work-up**: symptom and bother score, **3-day frequency–volume chart**, free flow and postvoid residual (PVR). A normal free flow lets an abnormal invasive flow be recognised as artefact.
- **Urinalysis before every invasive study**; postpone for symptomatic urinary tract infection (UTI). Commonest after-effect: dysuria.

Antibiotic prophylaxis (AUA / SUFU Best Practice Policy)

AUA/SUFU = American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction. The "index" patient needs **no routine prophylaxis** (level 1). Non-index patients get a **single dose within 1 h**: first choice oral trimethoprim–sulfamethoxazole; alternatives cephalosporin, amoxicillin–clavulanate, aminoglycoside with or without ampicillin, fluoroquinolone.

Prophylaxis YES	Prophylaxis NO
Neurogenic LUT dysfunction (NLUTD); bladder outlet obstruction (BOO) or raised PVR; age > 70; **asymptomatic bacteriuria**; immunosuppression (especially renal transplant); chronic catheters including clean intermittent catheterisation (CIC); high-risk orthopaedic implant	Diabetes; **recurrent UTI history**; recent hospitalisation; obesity; menopause; gender; prior urological surgery; cardiac valve disease (not routinely)

Trap: asymptomatic bacteriuria is treated here, unlike most urology.

3 · Non-invasive components

These cheap tests set targets for the invasive study.

- A 3-day bladder diary gives functional capacity, frequency and nocturnal output; it sets the cystometric target and shows if a flow is representative.
- P2 Uroflowmetry. — *Qmax, Qave, voiding efficiency; VV ≥150 mL; Siroky/Liverpool nomograms, corrected Qmax; bell, plateau (stricture), intermittent curves; cannot separate BOO from underactivity*

Not in the paper answer:

- Confirm a raised PVR with a repeat, especially if there was no true desire to void.
- Minimum voided volume is **VV 150 mL**, because maximum flow (Qmax) is linear above it and hyperbolic below.
- †Normal Qmax is > 15 mL/s in men (< 10 suggests obstruction) and > 20 mL/s in women.
- **PVR** is measured by scan or catheter (state which); there is no agreed threshold.

4 · Equipment, transducers and quality control

Many "abnormal" tracings are artefacts.

- Minimum: vesical pressure (Pves), abdominal pressure (Pabd), subtracted detrusor pressure (Pdet) and flow; electromyography (EMG) and urethral pressure are optional. Infuse saline or contrast, **not CO_2**.

Feature	External water-filled	Air-charged	Microtip
Reference level	**Upper border of symphysis pubis**; re-level on position change	None (no hydrostatic effect)	—
Problems	Air bubbles; motion and hydrostatic artefact; underdamped	Overdamped (signals > 3 Hz attenuated), so coughs are blunted	Expensive, reusable
Standards	Basis of most nomograms	No reference values	—

- Systems are **not interchangeable**: Pdet may differ by up to 10 cmH_2O. Patterns transfer, but absolute values do not.

> **KEY POINT**
> **Quality control** † (ICS Good Urodynamic Practice): zero to atmosphere; flush air; resting Pves and Pabd physiological and similar (Pdet near 0). **Cough test** before filling, at intervals and before voiding: equal spikes confirm transmission; a biphasic Pdet spike means re-flush or reposition. Artefacts: rectal contractions (unexplained fall in Pdet), catheter displacement, kinked tubing, stream hitting the funnel edge.

5 · Filling cystometry (CMG)

Filling cystometry (CMG) shows whether storage is at safe, low pressure.

- Filling to "permission to void" gives the **maximum cystometric capacity (MCC)**. **Pdet = Pves − Pabd**; rectal Pabd is preferred.
- Standard adult filling is **30–100 mL/min**; †the ICS physiological rate is < body weight (kg)/4 mL/min. Neurogenic bladders: physiological rate, body-warm saline (European Association of Urology, EAU). DO is commoner upright.
- A **normal ("stable")** bladder keeps Pdet near zero with no involuntary contractions.
- **Sensation** is the only subjective parameter: †first sensation, first desire, strong desire, urgency.

Detrusor overactivity (DO)

- DO means involuntary detrusor contractions (IDC) during filling, **idiopathic** or **neurogenic** (relevant neurological disorder). The label comes from history, as tracings look identical. "Hyperreflexia"/"instability" are obsolete.
- Absence does not exclude it: **up to 50%** of women with urgency urinary incontinence (UUI) show no DO. Some asymptomatic volunteers do show it.
- Confirm it is involuntary; record volume, pressure and leak; note phasic or terminal, suppressible or not.
- In BOO with overactive bladder (OAB), a **single terminal IDC** predicts storage-symptom resolution after transurethral resection of the prostate (TURP).
- In neurogenic DO (NDO), higher Pdetmax and longer IDCs predict upper-tract deterioration.
- **Cough-associated (stress-induced) DO** mimics stress urinary incontinence (SUI) but is detrusor in origin.

Compliance

- P3 Urinary bladder compliance — *ΔV/ΔPdet; < 20 mL/cmH_2O poor; absolute pressure matters more; causes; pitfalls (fast fill, low-amplitude IDC, VUR/diverticulum/outlet pop-off)*

Not in the paper answer:

- Storage pressure **> 40 cmH_2O** itself leads to upper-tract damage.
- Outlet pop-off hides poor compliance, which may appear only after a sling or artificial urinary sphincter (AUS); occlude the outlet.

6 · Leak point pressures

A low abdominal leak point means a weak sphincter; a high detrusor leak point endangers the kidneys.

- P1 Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications — *ALPP (low = weak sphincter) vs DLPP (high = upper-tract danger); technique from 150 mL, catheter out at capacity; VLPP < cough LPP; McGuire DLPP > 40 gives 85% hydronephrosis/VUR*

Not in the paper answer:

- Intrinsic sphincter deficiency (ISD) and hypermobility (Q-tip > 30°) coexist. With no hypermobility, the SUI must be ISD.
- Post-prostatectomy men often leak only with the catheter out. The ICS umbrella term is **urodynamic stress test**.
- **Occult SUI**: a continent woman with pelvic organ prolapse (POP) leaks only on prolapse reduction.
- After lowering a high DLPP, repeat the UDS.

Abdominal leak point pressure, ALPP (cmH_2O)	Inference (McGuire)
< 60	ISD

Abdominal leak point pressure, ALPP (cmH_2O)	Inference (McGuire)
60–90	Equivocal
≥ 90	Little or no ISD (hypermobility)

- ALPP poorly predicts surgical success.
- **DLPP > 40 cmH_2O** is not a validated safe cut-off (20 is more sensitive in children), so aim **as low as achievable**.
- Lower bladder pressure (or the outlet, e.g. sphincterotomy for DESD).

7 · Urethral pressure profilometry (UPP)

UPP measures urethral closure, but its value is doubted.

- Brown–Wickham: a perfused catheter is withdrawn. **Maximum urethral closure pressure (MUCP)** = maximum urethral pressure − intravesical pressure.
- Women: functional length ~3 cm, MUCP 40–60 cmH_2O.
- **MUCP ≤ 20 cmH_2O** means ISD ("type III" SUI) and predicts sling or suspension failure.
- ICS: utility unclear. EAU: do not use UPP or leak point pressure (LPP) to grade severity.

8 · Pressure–flow studies (PFS)

Only a pressure–flow study (PFS) tells obstruction from a weak bladder.

Pdet	Flow	Interpretation
Low/normal	High/normal	Unobstructed
High	Low	**BOO**
Low	Low	Detrusor underactivity (obstruction may be hidden)

- ICS: **detrusor underactivity** is reduced contraction strength or duration, so emptying is prolonged or incomplete. **Acontractile detrusor** ("neurogenic acontractile") replaces "areflexia".

Men — nomograms and indices

Abrams–Griffiths, Schäfer linear passive urethral resistance relation (LinPURR; grades obstruction and contractility) and the ICS provisional nomogram all come from TURP patients.

Index	Formula	Cut-offs
Bladder outlet obstruction index (BOOI)	PdetQmax − 2Qmax	**> 40 obstructed · 20–40 equivocal · < 20 unobstructed**
Bladder contractility index (BCI)	PdetQmax + 5Qmax	**> 150 strong · 100–150 normal · < 100 weak**

- BOOI ≥ 40 needs Pdet ≥ 40 even at zero flow, so a decompensated detrusor masks obstruction.
- **Stop tests** estimate isovolumetric pressure (Piso); in men Piso < 50 cmH_2O suggests underactivity.
- ~2/3 of men with benign prostatic obstruction (BPO) have DO; it resolves in half to two-thirds after relief.
- **Micturitional UPP** (Yalla) localises obstruction: a drop > 5 cmH_2O within the prostatic urethra.

- MCQ Bethanechol supersensitivity test (Lapides; Cannon's law): 2.5 mg subcutaneously; a rise **≥ 15 cmH_2O** at 100 mL is positive (neurogenic). Unreliable.

Women

- Women void at low pressure, so the **BOOI and ICS nomogram cannot be used**. Suspect BOO with incomplete emptying, recurrent UTI, prior anti-incontinence surgery or prolapse.

Method	Criterion	Comment
Cut-points	e.g. Qmax ≤ 12 mL/s and PdetQmax ≥ 25 cmH_2O (Defreitas); Chassagne ≤ 15 / ≥ 20	Anatomic obstruction only; overlap
Videourodynamics, VUDS (Nitti)	Radiographic obstruction between bladder neck and meatus with a sustained contraction of any magnitude	Localises; works despite weak detrusor
Blaivas–Groutz	Free Qmax vs Pdetmax; none/mild/moderate/severe	Two separate voids; **overdiagnoses**
Solomon–Greenwell	PdetQmax − 2.2Qmax: < 0 unlikely; > 18 almost certain	Includes functional obstruction

9 · EMG and sphincter coordination

A sphincter contracting during voiding raises bladder pressure.

- ICS standard: perineal surface electrodes; a concentric needle gives a better signal but hurts. Normally EMG quietens and urethral pressure falls **before** Pdet rises.
- The **guarding reflex** (EMG rise during an IDC) is normal, not dyssynergia.

	DESD	Dysfunctional voiding
Cause	**Suprasacral cord lesion** (the higher, the likelier)	Learned; no neurological lesion
Behaviour	Involuntary sphincter contraction during DO or voiding	Intermittent sphincter contraction during voiding
Significance	High pressure causes poor compliance and upper-tract risk	LUT symptoms; fluoroscopy improves accuracy

- **DISD** needs a suprasacral lesion above the T10–L1 sympathetic outflow and is seen on **VUDS only**.
- **Primary bladder neck obstruction (PBNO)** is invisible to EMG, so needs fluoroscopy.

10 · Videourodynamics

Fluoroscopy adds "where".

- P4 Video Urodynamics — *gold standard in NLUTD (EAU); only test for bladder neck/DISD/PBNO (types I–III); VUR pop-off; Nitti criteria; technique, screen only at key moments; indications*

Not in the paper answer:

- If VUDS is unavailable in NLUTD, do CMG with PFS (EAU).
- It confirms equivocal EMG, shows incontinence missed on examination, and gives DLPP in tetraplegics.
- Fluoroscopy is critical for PBNO types II and III.

11 · Ambulatory urodynamics

Natural filling avoids pump-filling artefacts.

- On **natural filling** some "poorly compliant" neuropathic bladders are normal, because rapid filling overstates pressure; DO is found more often.
- †Use it for symptoms unexplained by repeated conventional UDS. Penile cuff, wall thickness and near-infrared spectroscopy replace nothing.

12 · Indications by condition (AUA/SUFU, EAU)

Order UDS when the result will change treatment.

Female SUI

- Uncomplicated demonstrated SUI ("index" patient): UDS may be omitted (AUA/SUFU 2023; EAU: not routinely).
- Do UDS in non-index women: prior anti-incontinence or POP surgery, symptom–examination mismatch, voiding dysfunction, significant UUI, raised PVR, unconfirmed SUI, NLUTD.
- SUI not shown: repeat with catheter out; high-grade POP: reduce it.

Male LUTS

- No UDS is needed to start an α-blocker for lower urinary tract symptoms (LUTS).
- AUA/SUFU: do **PFS when obstruction matters, before invasive, morbid or irreversible treatment**; without obstruction, surgery does worse.
- EAU PFS triggers before invasive treatment: prior failed treatment; cannot void > 150 mL; **Qmax > 10 mL/s**; PVR > 300 mL; age > 80 or < 50.
- **Impaired compliance is the only absolute urodynamic indication to treat BOO.**
- OAB: no UDS initially; consider before invasive therapy.

Women with voiding symptoms; NLUTD

- PFS separates obstruction from impaired contractility; functional obstruction (PBNO, dysfunctional voiding) needs VUDS.
- NLUTD goal: **low-pressure filling**, even at the cost of catheterisation; only UDS separates DO from poor compliance.
- UDS is essential before treatment and in follow-up in SCI and myelomeningocele; helpful in BPO with Parkinson disease or MS; not usually first in post-stroke UUI.

13 · Landmark trials

UDS rarely changes outcome in uncomplicated SUI.

Trial	Design	Result	Take-home
ValUE	RCT, 630 women, uncomplicated SUI: office evaluation with or without UDS	Success ~77% both arms	Office evaluation alone non-inferior before SUI surgery
Dutch embedded RCT	Discordant UDS randomised: immediate mid-urethral sling vs UDS-tailored	Equal cure	UDS discordance need not change surgery
SISTEr	Burch vs pubovaginal sling	ALPP and DO did not predict outcome	UDS adds little in stress-predominant SUI

109 Neuromuscular Dysfunction of the Lower Urinary Tract

1 · Principles and classification

The lesion level predicts the bladder and sphincter pattern.

- Neuropathic lower urinary tract (LUT) dysfunction is largely **predictable from lesion site**; the acute picture may differ from the chronic one.
- The International Continence Society (ICS) uses **five parameters**: sensation; detrusor (normal, overactive, areflexic, impaired); compliance; smooth sphincter (synergic or dyssynergic); striated sphincter (synergic, dyssynergic, bradykinetic, impaired voluntary control, fixed tone).
- **Neuroplasticity** is reversible, then fixed. In bladder outlet obstruction (BOO), type III:I collagen rises (compliance falls), with denervation and nerve growth factor (NGF) afferent sensitisation; this causes detrusor overactivity (DO) persisting after de-obstruction.

Level	Typical disorders	Detrusor	Smooth sphincter	Striated sphincter
Suprapontine	Stroke, tumour, Parkinson disease (PD), dementia, cerebral palsy	Overactive	Synergic	Synergic (with or without impaired voluntary control; PD bradykinesia)
Multiple system atrophy (MSA)	—	Overactive, later impaired	**Open**	Denervated
Suprasacral cord	Spinal cord injury (SCI), multiple sclerosis (MS), transverse myelitis	Overactive	Synergic (dyssynergic if above T7)	**Dyssynergic**
Sacral cord / cauda equina	Sacral SCI, disc, myelodysplasia	Areflexic	Competent non-relaxing; may open	Fixed tone
Peripheral	Radical pelvic surgery, diabetes, tabes	Impaired / areflexic	Open (pelvic surgery)	Fixed tone / synergic

KEY POINT

Dyssynergia appears only when the lesion lies **between the pons and the sacral cord**. An open bladder neck marks autonomic (MSA, radical pelvic surgery, myelodysplasia) or sacral damage. PD and MSA share the detrusor pattern but **not the outlet**.

2 · Lesions at or above the brainstem

Brain lesions give an overactive bladder with a coordinated sphincter.

Stroke

- Urinary incontinence (UI) persisting 7 days predicts poor survival. The commonest type is impaired-awareness UI, then urgency UI (UUI).
- **Acutely** there is retention ("cerebral shock"). **Chronically** there is DO with intact sensation and **synergic sphincters**; UUI follows if voluntary sphincter control is lost (cortex or internal capsule lesion).

- **True detrusor–sphincter dyssynergia (DSD) does not occur. Pseudodyssynergia** is an electromyographic (EMG) flare from voluntary attempts to inhibit a contraction.
- In men with LUT symptoms (LUTS), do a full urodynamic study (UDS) before outlet surgery to exclude DO with impaired contractility.

Dementia, brain injury, tumour, others

- **Dementia**: anticholinergics worsen cognition and falls and raise dementia risk, so a **β3 agonist is strongly preferred** (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction, SUFU).
- **Brain injury** above the pontine micturition centre (PMC) gives DO with synergic sphincters; brainstem injury below the PMC can give DSD.
- **Brain tumour**: site (superior frontal lobe) matters, not histology; frontal lesions can cause retention.
- **Normal-pressure hydrocephalus** gives dementia, gait ataxia and UI; a shunt improves UUI.
- **Cerebral palsy** gives DO with synergic sphincters. About half of adults with LUTS have poor compliance or a detrusor leak point pressure (DLPP) ≥40 cm H_2O, so evaluate fully; conservative care succeeds in ~¾.

Parkinson disease vs multiple system atrophy

- **PD** results from nigral dopamine loss (loss of inhibitory D1, unopposed D2). LUTS occur in 35–70%, mainly storage. Sphincter **bradykinesia** mimics DSD.
- Treat with overactive bladder (OAB) drugs with low central nervous system (CNS) effect: mirabegron, trospium, darifenacin, extended-release fesoterodine. OnabotulinumtoxinA, posterior tibial nerve stimulation (PTNS) and sacral neuromodulation (SNM) work; subthalamic deep brain stimulation improves capacity.
- **Transurethral resection of the prostate (TURP) in PD is not contraindicated.** The old "post-TURP incontinence in PD" cases were MSA. Confirm PD, exclude MSA, and do videourodynamics first.
- **MSA** is a glial α-synucleinopathy. Loss in the intermediolateral columns and **Onuf nucleus** causes early severe LUTS and erectile dysfunction (ED).
- In MSA, DO comes early and poor emptying late. Treating DO worsens emptying, so add clean intermittent catheterisation (CIC); give desmopressin for nocturia. **Prostatectomy is hazardous.**

Feature	MSA	PD
Urinary symptoms / ED	Precede or accompany parkinsonism; ED near universal	Years after diagnosis; ED ~⅓
Dominant problem; PVR	UI; high postvoid residual (PVR)	Frequency/urgency; PVR normal
Bladder neck	**Open**	Synergic
Striated sphincter EMG	**Denervation**	Normal (bradykinesia)
Detrusor	DO, then areflexia	DO; areflexia uncommon
Outlet surgery	Worsens	Acceptable after work-up

3 · Spinal cord disease

Cord lesions add dyssynergia and silent upper-tract damage.

Multiple sclerosis

- LUTS occur in 50–90% and are the presenting complaint in up to 15%.
- On UDS, DO is found in ~⅔, DSD in 30–65% of those with DO, and areflexia in ~¼.
- Symptoms do not track the disease, so **baseline UDS is essential**.
- Keep management **reversible**, because the pattern changes: antimuscarinic with CIC; onabotulinumtoxinA; SNM in selected patients (MRI-compatible devices).
- Credé and Valsalva voiding are contraindicated. An indwelling catheter is the last resort, suprapubic if used.

Spinal cord injury

- P2 Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male — *spinal shock, suprasacral NDO + DSD, autonomic dysreflexia, CIC/toxin/sphincterotomy/Brindley ladder*

Not in the paper answer:

- State column and cord level separately: the sacral cord begins at vertebral column T12–L1, and the conus gives way to the cauda equina at ~L2.
- Stone risk is highest in the first 3 months (immobilisation hypercalciuria).
- Somatic completeness does not mean autonomic completeness.
- Spinal shock usually lasts **6–12 weeks**. Reflexes return together with the lower-limb deep tendon reflexes.
- Smooth sphincter dyssynergia occurs with cord lesions at **T7–T8** (column T6) or above.
- **Sacral lesions** give areflexia. Compliance is normal at first but may fall (decentralisation). The bladder neck does not relax (it may open later) and striated tone is fixed.
- Silent upper-tract damage can occur **without vesicoureteral reflux (VUR)**.

	Suprasacral (complete)	Sacral (complete)
Detrusor	Overactive	Areflexic
Compliance	Normal initially	Normal or high; may fall
Smooth sphincter	Synergic (dyssynergic above T7–T8)	Competent non-relaxing; may later open
Striated sphincter	Dyssynergic	Fixed tone, no voluntary control
Somatic signs	Spasticity, hyperreflexia	Flaccidity, reduced reflexes
Management	Lower pressure (drugs or toxin) and CIC	CIC and keep compliance safe

- **Neurology does not equal urodynamics.** Occult levels occur, and thoracolumbar (T10–L2 column) lesions are least predictable.
- **Manage by UDS, not by level**; one study may not suffice.
- Upper-tract risks are detrusor fibrosis, leak point pressure ≥75 cm H_2O and DO for >⅓ of filling. †DLPP >40 cm H_2O is the classic threshold.

Autonomic dysreflexia (AD)

- AD occurs in SCI **above T6** and may appear years later.
- Triggers are bladder or rectal distension, a blocked catheter, instrumentation, stones, sores, fracture and sex.

- Risk during UDS is highest with complete, cervical lesions, DSD and poor compliance.

ALGORITHM

1. †Sit the patient upright and loosen clothing.
2. **Remove the stimulus**: drain the bladder, unblock the catheter, empty the rectum.
3. If blood pressure stays high, apply nitroglycerin 2% paste above the lesion (first line).
4. If it still persists, give labetalol or intravenous nitroprusside.
5. For prophylaxis, give terazosin or prazosin (α1-blockers).
6. If intractable, consider sacral rhizotomy, sympathectomy or cordectomy.

- Nifedipine (10–20 mg sublingual) is now banned in many centres because of uncontrolled hypotension.

VUR and women with SCI

- A **chronic urethral catheter is worst** for the upper tracts. VUR artefactually enlarges capacity and lowers pressures on UDS.
- A destroyed urethra needs bladder neck closure with a suprapubic catheter, or diversion.
- Women have no collecting device: paraplegics do CIC, while tetraplegics often need indwelling catheters.

Other cord disorders

- **Transverse myelitis** causes retention, then DO with DSD; manage it as SCI.
- **Cervical myelopathy** gives DO with or without DSD, or underactivity (posterior funiculus), so UDS guides treatment.
- **Neurospinal dysraphism**: myelomeningocele is ~90% of spina bifida cystica, and its level correlates poorly with UDS.
- The typical adult has an **areflexic bladder and open bladder neck** with a fixed sphincter, so leaks at a fixed pressure; DSD occurs in 10–15%.
- Poor compliance, high pressure and DO predict renal decline.
- For stress UI, bulking is safer than a sling or artificial urinary sphincter (AUS) because it does not reduce compliance. Escalate from medical treatment and toxin to augmentation or diversion.
- **Tethered cord** presents after growth spurts; DO is commonest; urinary recovery lags behind pain and motor recovery.
- A **sensory neurogenic bladder** (tabes, B12 deficiency) has a large capacity. A **motor** one (polio) gives retention with intact sensation.

4 · Lesions distal to the cord

Root and nerve lesions give a weak bladder behind a fixed outlet.

- **Disc disease** (L4–L5, L5–S1) most often gives a **normally compliant areflexic bladder**, because the preganglionic injury is incomplete. Do **pre-laminectomy UDS** for clinical and medicolegal reasons.
- **Cauda equina syndrome** causes saddle anaesthesia, loss of voluntary sphincter control and sexual response, and an acontractile detrusor.
- **Decompression within 48 h** gives better urinary and rectal recovery. The sphincter recovers before the detrusor.

- **Radical pelvic surgery** (abdominoperineal resection, radical hysterectomy) gives impaired contractility, fixed striated tone, an **open non-functional bladder neck** and lower compliance. The patient leaks or retains; the bethanechol supersensitivity test is positive.
- After pelvic surgery, **use CIC and wait 6–12 months**; do **no prostatectomy unless BOO is proven**.
- **Herpes zoster** causes retention after the rash, resolving in 1–2 months. **Herpes simplex** sacral radiculitis with retention and cerebrospinal fluid (CSF) pleocytosis is **Elsberg syndrome**, which is transient.
- P2 Bladder dysfunction non genitourinary systemic disease — *diabetic cystopathy (reduced sensation, then raised capacity, reduced contractility and raised PVR; DO common; pressure–flow vs BOO; timed voiding), Guillain–Barré, lupus, B12*

Not in the paper answer:

- Diabetic "dyssynergia" is a straining artefact.
- **Guillain–Barré** causes retention; use CIC until recovery.

> **KEY POINT**
> Distal lesions hit the peripheral reflex arc, giving areflexia with a non-relaxing or fixed outlet. Compliance falls only when preganglionic parasympathetic motor fibres are damaged. Hence urgent decompression, waiting and CIC outrank outlet surgery.

5 · Specific functional and miscellaneous conditions

Several outlet problems look neurogenic but are not.

Detrusor–sphincter dyssynergia

- P1 Detrusor–external sphincter dyssynergia (DESD / DSD) — *lesion pons–sacral cord, Blaivas types 1–3, pseudodyssynergia, half of untreated men deteriorate, CIC first, sphincterotomy if CIC impossible*

Not in the paper answer:

- Blaivas type 1 (incomplete) can be observed unless there is VUR, hydronephrosis or AD; types 2–3 need treatment.
- Upper-tract complications are far less common in women than in men.
- Without a lesion, doubt DSD (the exception is Hinman syndrome in children). Misdiagnosing pseudodyssynergia leads to harmful treatment.

Functional outlet disorders

- **Dysfunctional voiding** (Hinman) is striated sphincter contraction during voiding without neurological disease, proven on pressure–flow–EMG.
- **Non-neurogenic chronic retention** is PVR ≥**300 mL** persisting ≥6 months on ≥2 occasions. High-risk patients (hydronephrosis, chronic kidney disease, urinary tract infection [UTI]) need drainage; suprapubic is better than urethral long term.
- **Primary bladder neck obstruction** affects young and middle-aged men with a normal prostate. Videourodynamics localises the BOO to the bladder neck.
- α-blockers help some, but **bladder neck incision is definitive**. A "trapped prostate" (Turner-Warwick) needs a small TURP including the bladder neck.
- **Female BOO** is rare. Groutz criteria: free maximum flow (Qmax) ≤12 mL/s and detrusor pressure at Qmax ≥20 cm H_2O. Bladder neck surgery risks sphincteric UI.
- **Fowler syndrome** is retention in young women (capacity >1 L), often with polycystic ovaries. Needle EMG shows **complex repetitive discharges and decelerating bursts**; urethral

closure pressure is >100 cm H_2O.

- **SNM is the treatment for Fowler syndrome** (~70% succeed), and the syndrome predicts SNM response.

Other conditions

- **Myasthenia** raises incontinence after prostatectomy. **Isaacs syndrome** causes sphincter spasm, treated by suprapubic drainage and plasmapheresis. **Wernicke** encephalopathy responds to thiamine.
- Myelopathies (tropical spastic, schistosomal, syringomyelia) give DO with DSD; cytomegalovirus and Lyme disease give areflexia. **Lupus cystitis** is treated with steroids.
- After **radiation**, early storage symptoms recover by 6 months. A **defunctionalised bladder** is rehabilitated by cycling; transplant is possible without augmentation.

6 · Evaluation and risk stratification

Patients are ranked by upper-tract risk.

- Guidelines: American Urological Association (AUA)/SUFU neurogenic LUT dysfunction (NLUTD) 2021; European Association of Urology (EAU) Neuro-urology 2022. **Primary goal: prevent renal damage.**
- Everyone gets history, examination (cognition, **hand dexterity**, spasticity), urinalysis, PVR and diaries.
- **Defer full assessment during spinal shock or acute brain injury**, and give no irreversible therapy then.

Risk (AUA/SUFU)	Typical	Defining findings
Low (all)	Suprapontine or uncomplicated peripheral	Spontaneous voiding, low PVR, stable; normal imaging and renal function
Unknown	Infrapontine–suprasacral; sacral/infrasacral	Storage and voiding symptoms, raised PVR, recurrent UTI; not yet assessed, so do UDS, upper-tract imaging and renal function
Moderate	—	Neurogenic retention; DO with incomplete emptying; normal tracts
High (any one)	—	Poor compliance, raised storage pressure, DO with DSD, VUR; hydroureteronephrosis, scarring, large stones; abnormal renal function

KEY POINT

Low risk needs no routine imaging, renal function or UDS. DO alone fits low risk (suprapontine, synergic); high risk means DO with infrapontine pressure consequences. Risk is judged by **pressure, compliance and upper tracts**, not by diagnosis.

7 · Management

Start with the simplest, most reversible step.

- **Goals**: protect the upper tracts; control infection; achieve low-pressure storage and emptying; continence; avoid catheter or stoma if possible; social acceptability.

ALGORITHM

1. Aim for low-pressure storage with regular emptying: CIC with an antimuscarinic or β3 agonist.
2. If refractory, give intradetrusor onabotulinumtoxinA.
3. In selected patients, try neuromodulation.
4. If these fail, do augmentation or continent diversion.
5. As a last resort, do bladder neck closure with a suprapubic catheter, ileovesicostomy or incontinent diversion.

- **CIC is the gold standard**; keep volumes **< 500 mL**. Credé and Valsalva are not recommended.
- CIC beats an indwelling urethral catheter (less pyelonephritis, epididymo-orchitis and urosepsis). For a chronic catheter, **suprapubic is better than urethral**.
- For refractory neurogenic DO (NDO), use **intradetrusor onabotulinumtoxinA** (~60% continue long term) and warn about CIC. Intrasphincteric toxin is not routine.
- The Brindley stimulator (deafferentation with an anterior root stimulator) is used in complete SCI; the rhizotomy helps even with the stimulator off *(see Ch 115)*.
- PTNS suits patients who void and have OAB (MS, PD, stroke); SNM suits NLUTD without SCI or spina bifida.
- Retropubic bladder neck closure is better than perineal.
- **UTI**: treat bacteriuria only if symptomatic, correct stones and residuals, and give **no prophylaxis** for CIC or indwelling catheters. Refractory cases need upper-tract imaging, cystoscopy and UDS.
- **Bladder cancer** incidence is comparable to the general population, but it is often muscle-invasive at diagnosis; catheters and UTI are the risk.
- **AUA/SUFU 2021: no routine surveillance cystoscopy**; scope for haematuria.

8 · Follow-up

Follow-up is lifelong, because upper-tract damage is silent.

Body	Schedule
AUA/SUFU 2021 (risk-based)	Low risk: none routine. Moderate/high: yearly history, examination and renal function; upper-tract imaging every 1–2 years; repeat UDS for change; after bowel reconstruction, yearly electrolytes, acid–base and imaging
EAU	Specialist review every 1–2 years; ultrasound and residual 6-monthly; dipstick 2-monthly; yearly examination and bloods
American Paraplegia Society (SCI)	Annual for 5–10 years, then alternate years; renogram: a >20% fall in renal plasma flow means investigate; annual cystoscopy with indwelling catheter (**not endorsed** by AUA/SUFU)

110 Overactive Bladder

1 · Definitions and terminology

OAB is a symptom diagnosis built on urgency.

- P1 P2 Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management; Overactive bladder and its drug therapy — *ICS definition, OAB vs DO, afferent/urothelial/myogenic/central mechanisms, behavioural therapy, drugs, PTNS/toxin/SNM*

Not in the paper answer:

- There is no true animal model of OAB, because animals cannot report urgency.
- Micromotions are normal localised contractions, felt as filling without a pressure rise.
- In overactive bladder (OAB), **urgency is the defining symptom**. "Urge" incontinence became "urgency" incontinence to tie it to that sensation.
- OAB is a symptom syndrome. Detrusor overactivity (DO) is a urodynamic observation, which may be asymptomatic or artefact.

Term	ICS definition (all patient-reported)
Increased daytime frequency	Patient considers they void too often by day; upper normal 7 voids
Nocturia	Waking ≥1 time to void; each void preceded and followed by sleep
Urgency	Sudden, compelling desire to void, difficult to defer
UUI	Involuntary leakage associated with urgency

- **Bother**: **urgency urinary incontinence (UUI) is the most bothersome lower urinary tract symptom (LUTS) in both sexes**. Frequency bothers at ≥**8** daytime voids, nocturia at ≥**2**.
- **Isolated nocturia** usually means nocturnal polyuria or poor sleep. †Nocturnal polyuria (International Continence Society, ICS) is night urine >33% of 24-h output in older adults.
- **Mixed urinary incontinence (MUI)** is stress UI (SUI) with UUI. SUI with "OAB dry" is **not** MUI. Urinary and faecal leakage together is **double incontinence**.
- OAB wet carries more psychological burden and more DO (almost always present in men).

OAB versus bladder pain syndrome (BPS)

BPS requires bladder pain, pressure or discomfort with ≥1 urinary symptom; pain is excluded from OAB. Overlap is large (shared afferent hypersensitivity).

Feature	OAB	BPS
Primary fear	**Leakage**	**Pain**
Filling sensation	Phasic urgency waves	Steady, accelerating pain
Voided volumes	Variable	More consistent
Deferral	Limited, because of leak risk	Possible, at cost of pain
Site	Perineum/penile base, urethra/vagina	Suprapubic, with or without perineal
UUI	Yes	No

2 · Epidemiology and burden

OAB is common, under-reported and poorly persisted with.

- Adult OAB affects ≈ **12–29%**: the loosest definition (EpiLUTS) gives the highest figure, strict ICS (EPIC) the lowest.
- Women have OAB 1.5–2× and UUI 2–3× as often as men (NOBLE UUI 7.6% vs 2.6%), so the sex gap is in **wet** OAB.
- Beyond 75, ~40% of men and ~⅓ of women are affected; prevalence is highest in Black people.
- OAB dry progresses to UUI at ~1%/year; remission is ~6%/year.
- Fewer than half tell a clinician. ~**92%** stop the first antimuscarinic within 2 years, because of **unmet expectations** more than side effects, and side effects more than cost.
- Third-line: onabotulinumtoxinA 100 U is most cost-effective, sacral nerve stimulation (SNS) costliest.

3 · Pathophysiology

Urgency arises when too much bladder signal reaches awareness.

- **DO is no longer the sine qua non**: only ~half with DO have urgency, and **44–69%** with urgency show DO.
- The **periaqueductal grey (PAG) and pontine micturition centre (PMC) form the main regulatory level**; the forebrain (insula, prefrontal cortex) handles sensation and volition.
- Silent C fibres awaken. Urothelium-based and myogenic (exaggerated **micromotions**) hypotheses, bowel cross-talk and failed central nervous system (CNS) filtering of "afferent noise": see pointer.

> **KEY POINT**
> Final common pathway: more afferent signal reaches awareness. The bladder or a neighbour turns it up, or the CNS fails to turn it down. Hence DO and obstruction correlate only loosely with OAB.

4 · Aetiology and risk factors

Ageing is the main driver.

- Risks are **ageing** (clearest), female sex, metabolic syndrome, bowel and hormonal factors, depression and anxiety.
- **Bladder outlet obstruction (BOO)** is the standard animal model, but only an **indirect** cause in humans: OAB is as common in women, in whom BOO is rare. Both are probably parallel results of ageing.
- **Behaviour** such as "just in case" voiding and coping (toilet mapping, fluid restriction) sustains a negative bladder perception.

5 · Clinical assessment

A short work-up is enough to start treatment.

- The **minimum** (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction, AUA/SUFU; European Association of Urology, EAU, agrees) is **history, examination and urinalysis**, followed by empirical treatment. The EAU adds questionnaires and a diary.
- **History**: bother; SUI; voiding symptoms; haematuria; fluids and caffeine; occult neurology (new OAB with erectile dysfunction or tremor); pelvic surgery or radiotherapy; bowel (faecal

incontinence in ~**50%** of women with OAB); drugs; polyuria (diabetes, sleep apnoea, lithium).

Drug class	Relative contraindications
Antimuscarinics	Uncontrolled closed-angle glaucoma; cognitive impairment; retention; impaired gastric emptying
β3-adrenergic agonists	Uncontrolled hypertension; arrhythmia

- **Examination**: frailty (Timed Up and Go), cognition, leg oedema. The **pelvic and genital examination is most important**: prolapse, atrophy, stress test; digital rectal examination (DRE) in men; perineal sensation and anal tone.
- **Urinalysis is mandatory**; culture if infection is suspected; microhaematuria needs full evaluation.
- **Postvoid residual (PVR)** is not needed before conservative or oral therapy if uncomplicated. Measure it with retention risk, voiding symptoms (men), prior outlet surgery or neurology.
- Use a **diary** for 3–7 days (minimum 24–48 h). **Maximum voided volume is the functional capacity.** Sensation grade 0 is a "just in case" void; 2–3 is urgency; 4 is urgency with leak.
- Questionnaires: ICIQ-OAB with ICIQ-OABqol; AUA/SUFU suggest UDI-6, IIQ or OAB-q. MCQ OAB-V8 is a screening tool, not a severity score.

6 · Urodynamics (UDS)

Keep urodynamics for invasive decisions.

- UDS is **not** in the initial work-up (nor cystoscopy, ultrasound). It is **indicated** when the result decides invasive or irreversible treatment, after multiple failures, or with suspected BOO (OAB after an outlet procedure). In the index woman it improves satisfaction, not outcome.
- Findings are **DO** or **increased sensation**. Stop antimuscarinics to show DO; continue them to explain refractory symptoms.
- DO is found in OAB dry in 60–70% of men and 40–50% of women. In OAB wet it is found in **>90% of men** but is absent in 30–40% of women (SUI misread as urgency).
- DO correlates **best with UUI and worst with nocturia**.
- Provoke DO (running water, standing); if low-volume DO masks SUI, refill supine.
- **Assess emptying**: each **10 mL** of PVR raises the risk of catheterisation after onabotulinumtoxinA by **27%**. DO persists in up to **60%** after transurethral resection of the prostate (TURP).

7 · Management

Treatment follows the patient's goals.

- OAB is benign, so **no treatment is acceptable**. Incontinence-associated dermatitis justifies more aggressive care; immobility-driven leakage needs mobility treatment.
- Set **realistic expectations**: improvement, not cure. AUA/SUFU lines are **not a compulsory sequence**; allow **4–12 weeks** per therapy.

ALGORITHM

1. Confirm OAB and exclude mimics.
2. Educate, then start conservative measures.
3. If not enough, add drugs.
4. If drugs fail, offer tibial nerve stimulation, sacral neuromodulation or onabotulinumtoxinA.
5. Rarely, do augmentation or diversion.
6. Reassess at every step.

- Reducing fluid by ~**25%** helps. Cutting caffeine improves urgency and frequency, but not consistently UUI.
- Pelvic floor muscle training (PFMT) with bladder training reduces leaks by **50–80%**.
- **Weight loss** reduces SUI and UUI (†PRIDE randomised trial, obese women). After bariatric surgery, each 5% lost lowers incontinence risk by ~7–8%.
- Drugs *(see Ch 113)*: response is idiosyncratic, so switch or combine before invasive work-up.
- In **dementia**, antimuscarinic use >**3 months** raises risk by 10–65%, so prefer a β3 agonist.
- In benign prostatic hyperplasia (BPH) with OAB, an α1-blocker plus antimuscarinic works; retention occurs in ~3%, so take care if PVR is raised.

Third-line *(see Ch 115)*

Frailty predicts functional decline and 1-year morbidity of up to 50%, even after minor procedures.

	Tibial nerve stimulation	OnabotulinumtoxinA	SNS
Efficacy	60–80% respond	UUI reduced 59%; ~¼ dry	5-year success 70–80%
Drawback	12 weekly visits; response by weeks 8–10	Urinary tract infection (UTI) ~20%; catheterisation 0–12% in trials, real-world ~20% women, ≤35% men	Revision >30% at 5 years; battery
Risk	Lowest	Moderate	Highest

- **International Consultation on Incontinence (ICI) pathways**: in men, treat the outlet with the bladder (α-blocker if BOO). In women, treat the most bothersome MUI component first. In the frail elderly, put comorbidity and carer goals first, then try an antimuscarinic cautiously.

KEY POINT

Escalate by invasiveness and reversibility, not severity alone. OAB is benign, so patient goals, frailty and PVR decide how far to climb.

111 Underactive Detrusor

- P2 P4 Management of detrusor underactivity; Detrussor underactivity. — *ICS definition, causes (neurogenic/myogenic/drugs/functional), BCI, management table: CISC mainstay, α-blockers, bethanechol not routine, SNM, Brindley, outlet surgery, diversion, investigational options*

Not in the paper answer:

- Significant PVR proposal (over 40% of capacity); DHIC; multisystem atrophy 52–95%.
- PIP and DECO thresholds; ambulatory UDS finds contraction in over four-fifths.
- Poor surgical outcome: voiding pressure under 45 cm H_2O, age over 80, PVR over 1500 mL.
- Sphincter toxin 50–100 U; no bladder neck incision in women.

1 · Definitions and symptoms

Why this matters: drugs and surgery lower outlet resistance but cannot strengthen the detrusor.

- **Detrusor underactivity (DUA)** is managed mainly by drainage with intermittent self-catheterisation (ISC).
- **DUA** (International Continence Society [ICS] 2002, urodynamic) is a contraction of **reduced strength and/or duration**, causing prolonged or incomplete emptying. **No thresholds** are given.
- An **acontractile detrusor** does not contract. A **"bashful bladder"** fails to void only during the urodynamic study (UDS), and often in public toilets.
- **Underactive bladder (UAB)** (ICS 2018) is the symptom set: slow stream, hesitancy, straining, with or without incomplete emptying, sometimes with storage symptoms.
- UAB has **no pathognomonic symptom** and overlaps with overactive bladder (OAB) and bladder outlet obstruction (BOO).
- A significant postvoid residual (PVR) is not agreed; one proposal is over **40%** of functional capacity (voided plus PVR).

Phase	Symptoms
Voiding	Weak stream, intermittency, hesitancy, straining
Post-micturition	Incomplete emptying
Storage	Frequency and nocturia, **or** reduced desire with infrequent voiding
Very large PVR	Incontinence, especially in sleep

> **KEY POINT**
> **Sensation decides the picture.** Absent sensation makes an infrequent voider; with poor emptying, overflow is worse at night because the urethra relaxes in sleep. Preserved sensation makes a frequent voider with incomplete emptying and nocturia. DUA may be asymptomatic.

2 · Epidemiology

Why this matters: DUA is common in the elderly and coexists with other dysfunction.

- Population prevalence is **unknown** (no non-invasive marker). BOO is rare in women (2.7–8% of referrals), so low flow with high PVR suggests DUA.
- UDS series find DUA in men under 50 in **9–28%**, men over 70 in ~**48%** and women in 12–45%.

- It peaks in institutionalised elderly people, often as **detrusor hyperactivity with impaired contractility (DHIC)**: an involuntary contraction emptying under half the bladder.
- Half of elderly men also have detrusor overactivity (DO) or BOO, and ~¾ of women DO or stress urinary incontinence (SUI). So do **UDS before lower urinary tract symptom (LUTS) surgery in the elderly**.
- Non-neurogenic male DUA shows **minimal progression** over 10 years.

3 · Aetiopathogenesis

Why this matters: each cause breaks one link of the voiding reflex.

Confirmed causes are neurological disease and diabetes; probable ones are BOO, ageing and atherosclerosis.

Mechanism	Site	Result
Myogenic	Myocyte, extracellular matrix (ECM)	Weak contraction despite normal drive
Efferent	Sacral cord, roots, pelvic nerves	Impaired activation
Afferent	Peripheral afferents, spinal columns	**Early termination** of the voiding reflex
Brain	Pontine micturition centre (PMC), periaqueductal grey (PAG), limbic, prefrontal	Failed integration

- **Myogenic**: ECM fibrosis gives the Elbadawi **degeneration pattern** on electron microscopy. Asymptomatic elderly women show it too, so ageing contributes.
- **Afferent**: bladder afferents (volume, contraction) and urethral afferents (flow) sustain the reflex; sensation falls with age.
- **"Stiff balloon" hypothesis (Smith)**: fibrosis makes wall stress rise abruptly, so sensation is late and then urgent. After a small void the stress collapses, afferent drive fades and the contraction stops early, leaving a PVR.
- Other causes: tabes, spinal stenosis, pelvic fracture, radiotherapy. **Drug causes are reversible**: anticholinergics (including antihistamines, antipsychotics, tricyclics, antiparkinsonian drugs) and **opioids**.
- †**Fowler syndrome**: young women in retention with a non-relaxing striated sphincter on electromyography (EMG); it responds to sacral neuromodulation.
- **BOO**: animal hypertrophy leads to **decompensation**, perhaps via wall tension (Laplace), ischaemia–reperfusion and denervation.
- Yet in men followed over 10 years, detrusor pressure at maximum flow (Pdet@Qmax) did not fall, and most men in acute retention have preserved detrusor function.
- **Diabetes** follows a **temporal theory**: early osmotic diuresis causes hypertrophy and storage symptoms; late oxidative damage impairs sensation and emptying.

Condition	DUA	Key point
Stroke	~Half retain acutely	"Cerebral shock"; chronic sequel is DO
Parkinson disease	<20%	DO commoner; DUA drug-related
Multisystem atrophy	**52–95%**	Efferent parasympathetic atrophy; mistaken for Parkinson
Multiple sclerosis	~20%	Lumbosacral plaques

Condition	DUA	Key point
Radical pelvic surgery	Hysterectomy: ~72% LUT dysfunction	Nerve-sparing reduces it; many recover by 1 year
Guillain–Barré, zoster	—	Reversible

4 · Diagnosis

Why this matters: only a pressure-flow study proves a weak detrusor.

- **Invasive UDS is the only accepted method**. There are no consensus criteria, and nearly all measure strength only.
- Typically maximum flow (Qmax) and Pdet@Qmax are both low. Pitfalls: the **bladder outlet relation** (Hill equation) makes pressure and flow inverse, so Pdet@Qmax underestimates strength; and outlet resistance is ignored. Hence isovolumetric estimates.

Measure	Formula	Thresholds / note
Bladder contractility index (BCI) (Abrams)	Pdet@Qmax + 5Qmax	>150 strong; 100–150 normal; **<100 weak**; male-derived
Projected isovolumetric pressure (PIP) (Schäfer)	Pdet@Qmax + K·Qmax (K 5 in men, 1 in older women)	<50 very weak; 50–100 weak
Detrusor contraction coefficient (DECO)	BCI/100	<1 weak

- †The **BOO index** is Pdet@Qmax − 2Qmax: over 40 obstructed, under 20 unobstructed.
- **Stop tests**: a voluntary stop reads ~20 cm H_2O lower because of reflex inhibition. Continuous occlusion is most reliable but painful (research only).
- Falling shortening velocity is an early marker, preceding a fall in Watts factor.
- If the patient fails to void on UDS, do **ambulatory UDS**: over four-fifths of apparently acontractile patients contract.

5 · Management

Why this matters: safe drainage protects the kidneys; no drug reliably helps.

ALGORITHM

1. Evaluate: diary, digital rectal examination, urinalysis, flow, PVR.
2. If there is a neurological deficit, then do spinal MRI.
3. Stop offending drugs; treat constipation.
4. Timed or double voiding.
5. Lower outlet resistance where relevant.
6. If PVR is high, then ISC; if ISC is impossible, then a suprapubic catheter.
7. Selected patients: sacral neuromodulation (SNM), outlet surgery, diversion.

Aim	Routine	Experimental	Not recommended
Drainage	Timed/double voiding; catheterisation; diversion	—	Credé, Valsalva, reflex voiding
Lower outlet resistance	Pelvic floor biofeedback; α-blockers; outlet surgery	Sphincter botulinum toxin	—

Aim	Routine	Experimental	Not recommended
Improve detrusor function	SNM; anterior root stimulator	Intravesical prostanoids; intravesical electrotherapy; myoplasty	**Muscarinic agonists; anticholinesterases**

- **Credé or Valsalva** is used only with an incompetent (neurogenic) sphincter; otherwise it drives reflux into ureters and prostate.
- **ISC** causes urethral bleeding in ~⅓. †Keep catheterised volumes below ~500 mL. Suprapubic is the best long-term indwelling route.
- **Bethanechol and distigmine** cause bronchospasm, diarrhoea, salivation and **rare fatal cardiac arrest**, and are useless with fibrosis.
- An α-blocker plus a parasympathomimetic improved flow in one single-blind randomised trial. Prostaglandin E_2 (PGE_2: contracts detrusor, relaxes urethra), ASP8302 and TAC-302 are investigational *(see Ch 113)*.
- **SNM** inhibits urethral afferents that block bladder afferents. It works best with a sphincter or pelvic floor disorder; it may help DHIC, where treating DO worsens DUA and vice versa *(see Ch 115)*.
- **Brindley** stimulator: for complete spinal cord injury (SCI); it needs efferents and muscle and is combined with rhizotomy.
- **Intravesical electrotherapy (Katona)** stimulates Aδ afferents, mainly in children; the only sham-controlled trial was negative.

KEY POINT

Every therapy needs a surviving link: Brindley needs efferents and muscle; neuromodulation and electrotherapy need afferents and muscle. **Preserved sensation predicts response** to bethanechol, toxin and electrotherapy.

- **Sphincter toxin**: 50–100 U cystoscopically (men) or periurethrally (women). About half of idiopathic cases improve, best with poor relaxation and good sensation. It is short-lived and unlicensed.

	DUA, low PVR, symptomatic	DUA, catheter-dependent
Reasoning	Emptying at low Pdet, so BOO is unlikely to matter	Lowering resistance may allow voiding
Evidence	Transurethral resection of the prostate (TURP) at ~11 years: no benefit over conservative care	Flow rarely normalises, but many void
Verdict	**Surgery unlikely to help**	Consider in young, fit patients

- Outcome in retention is poor with voiding pressure under **45** cm H_2O, age over **80** or PVR over **1500 mL**.
- In women, avoid bladder neck incision (incontinence, stenosis).
- **Diversion**: a continent catheterisable stoma suits young women with idiopathic retention refractory to SNM; an ileal conduit treats renal deterioration from intramural ureteric obstruction.
- Latissimus dorsi myoplasty let most acontractile patients void, with complications in ⅓. Reduction cystoplasty is abandoned.

113 Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

1 · Framework

Why this matters: it matches each drug to the failure.

In the Wein functional scheme *(see Ch 104)*, **storage failure** is treated by relaxing the bladder or raising outlet resistance; **emptying failure** by increasing detrusor contraction (weak tools) or lower outlet resistance. Most lower urinary tract (LUT) drugs fail on **selectivity (therapeutic window)**: the same receptor in heart and vessels caps the dose.

2 · Antimuscarinics and β3-agonists

Why this matters: first-line overactive bladder (OAB) drugs; side effects and age drive choice.

- P1 P3 Compare and contrast various detrusor relaxants in management of OAB; OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron — *M3 vs M2, agent table with doses, β3/cAMP, mirabegron ≥180/110 and CYP2D6 (metoprolol, desipramine, digoxin), vibegron 75 mg, dementia risk, SYNERGY/BESIDE*

Not in the paper answer:

- Transdermal oxybutynin; Beers avoidance of oxybutynin; myasthenia and severe colitis as contraindications.
- MILAI II add-on to mirabegron; combination doubles constipation.
- †Antimuscarinics act mainly in **storage** (less urgency, more capacity, damped afferents), not on the voiding contraction, so retention is uncommon.
- **Oxybutynin** causes most dry mouth (metabolite N-desethyloxybutynin); transdermal use bypasses first-pass, so less. It has most central nervous system (CNS) penetration: avoid in the elderly (Beers).
- Tolterodine: 2 mg bd or extended-release (ER) 4 mg od. **Darifenacin**: most M3-selective; constipation.
- **Trospium** is a quaternary amine, renally excreted. Propiverine 20 mg (mixed action); imidafenacin 0.1–0.2 mg bd (Japan).
- †**Side effects by receptor**: dry mouth, constipation, blurred vision (M3); tachycardia (M2); cognition (**M1**).
- †**Contraindications**: **uncontrolled narrow-angle glaucoma**, urinary or gastric retention, myasthenia gravis, severe ulcerative colitis or toxic megacolon.
- †Elderly: cumulative **anticholinergic burden** favours a β3-agonist or non-CNS-penetrant agent. Men with bladder outlet obstruction (BOO) are safe if postvoid residual (PVR) is low; check it before and during.
- Mirabegron goes from 25 to 50 mg od; avoid in severe uncontrolled hypertension (≥**180/110** mm Hg). Vibegron needs no titration and had no blood pressure (BP) effect.

ALGORITHM

1. †OAB: behavioural therapy, then an antimuscarinic *or* a β3-agonist; if elderly or at cognitive risk, then choose the β3-agonist first.
2. If the response is poor, then switch or combine.
3. If refractory, then onabotA 100 U, sacral neuromodulation (SNM) or percutaneous tibial nerve stimulation (PTNS).
4. Last resort: augmentation cystoplasty.

3 · Combination therapy

Why this matters: the next step when one drug fails.

Direction	Evidence	Agents
Mirabegron added to failing antimuscarinic	SYNERGY, SYNERGY II, BESIDE, MILAI	**Solifenacin only**
Antimuscarinic added to failing mirabegron	MILAI II (52 weeks)	Solifenacin 5, propiverine 20, imidafenacin 0.2, tolterodine ER 4 mg
Vibegron plus antimuscarinic	Phase IIb (vibegron 100 plus tolterodine 4 mg)	Extrapolated; no trial after monotherapy failure

KEY POINT

Combination beats solifenacin 5 mg and either drug alone. In non-responders it **equals** solifenacin 10 mg with less dry mouth: it gains **tolerability, not efficacy**, over dose escalation. Heart rate, BP, QT and PVR are unchanged; constipation roughly doubles. Benefit holds at age ≥65 or ≥75.

- **α-blocker plus desmopressin** suits refractory nocturia, as ≈30% of men ≥60 with benign prostatic hyperplasia (BPH) have nocturnal polyuria; evidence is uncontrolled.
- **Vaginal oestrogen plus antimuscarinic**: additive benefit **not established** (simultaneous randomised controlled trials [RCTs] null; vaginal estradiol ≈ oxybutynin). †Systemic oestrogen worsens urinary incontinence (UI): do not use.

4 · Intradetrusor onabotulinumtoxinA (onabotA)

Why this matters: the main third-line option.

- P1 Botulinum toxin in urology: mechanism, indications and doses — *SNAP-25 cleavage, 100 U/20 sites OAB (EMBARK), 200 U/30 sites NDO (DIGNITY), PVR at 2 weeks, UTI, myasthenia*

Not in the paper answer:

- Repeat interval, aminoglycoside caution; rimabotulinumtoxinB too short-acting.
- The dose is **100 U** for idiopathic OAB and **200 U** for neurogenic detrusor overactivity (NDO).
- DIGNITY: equal efficacy in anticholinergic users and non-users. EMBARK: refractory patients only.
- †Retention needing clean intermittent catheterisation (CIC) is higher in non-catheterising NDO, as in multiple sclerosis (MS); patients must accept CIC first.
- Commonest adverse event (AE): urinary tract infection (UTI). Effect lasts ≈6–9 months; repeat ≥12 weeks apart. Caution with aminoglycosides.

5 · Nocturia: desmopressin †

Why this matters: it treats nocturnal polyuria but risks hyponatraemia.

- P3 Desmopressin (DDAVP) — *V2 agonist for nocturnal polyuria; hyponatraemia risk age ≥65; sodium at baseline, 1 week, 1 month; contraindications incl. loop diuretics*
- Low-dose sublingual: women 25 µg, men 50 µg; key risk rises at age ≥**65**.

6 · Increasing outlet resistance (SUI)

Why this matters: no drug is standard for stress urinary incontinence (SUI).

†Pelvic floor muscle training (PFMT) and surgery are first-line. Systemic α-agonists lack urethral selectivity, so BP rises before closure improves.

Drug	Mechanism	Status
Duloxetine †	Serotonin–noradrenaline reuptake inhibitor (SNRI) at **Onuf's nucleus**; raises pudendal tone	40 mg bd; nausea causes discontinuation; Europe, not USA
Phenylpropanolamine, midodrine, norepinephrine	Systemic α-agonists	Weak evidence; haemorrhagic stroke led the Food and Drug Administration (FDA) to ban phenylpropanolamine
Imipramine †	Anticholinergic plus α-like outlet effect	Mixed UI, enuresis; orthostatic hypotension, arrhythmia; caution in elderly
Clenbuterol	β-agonist	Small flawed RCT; tremor; abandoned
PSD503 vaginal phenylephrine gel	Local α-agonist	Poor acceptability; abandoned

7 · Facilitating emptying

Why this matters: stopping culprit drugs helps more than adding any.

- †**Bethanechol** has **no convincing controlled evidence** in detrusor underactivity and causes cholinergic AEs; contraindicated in asthma, peptic ulcer, bradycardia and mechanical obstruction.
- †**α-blockers** help in BPH, primary bladder neck obstruction and neurogenic LUT dysfunction, and aid trial without catheter.
- †Striated sphincter: baclofen, dantrolene (limited); sphincteric onabotA.
- †**Drugs that cause retention**: anticholinergics (antihistamines, tricyclics, antipsychotics, antiparkinsonian), pseudoephedrine, opioids, benzodiazepines, anaesthesia and epidurals. Ask in every retention.

8 · Failed drugs

Why this matters: they show the selectivity problem.

Agent	Indication	Reason for failure
Vitamin D3, elocalcitol	Urgency UI (UUI), OAB	No urodynamic effect
Calcium antagonists	OAB	No effect

Agent	Indication	Reason for failure
K^+ channel openers (ZD0947, flupirtine)	OAB	Dose capped by heart; flupirtine hepatotoxic
URO-902 (big potassium [BK] channel gene therapy)	OAB	Phase 1 only
RimabotulinumtoxinB	Toxin A-refractory DO	Short duration (≤3 months)
Intraprostatic onabotA, PRX302, NX-1207	BPH lower urinary tract symptoms	No better than sham injection

9 · Landmark trials

Why this matters: they set today's combination and toxin doses.

Trial	Design	Result	Take-home
SYNERGY / II	Phase III, 12 weeks / 12 months	Solifenacin 5 plus mirabegron 50 better than monotherapy for UI	Effective, safe long-term
BESIDE	Solifenacin 5 non-responders	Add-on better than solifenacin 5, equal to solifenacin 10	Alternative to dose escalation
MILAI II	52 weeks, mirabegron non-responders	Four antimuscarinic add-ons safe, effective	Add antimuscarinic to mirabegron
DIGNITY	Pooled RCTs, NDO	OnabotA 200 U equal in anticholinergic users/non-users	Efficacy independent of anticholinergic
EMBARK	RCTs, refractory OAB	OnabotA 100 U effective	Standard idiopathic OAB dose

115 Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

- P4 P3 Neuromodulation for voiding dysfunction, including sacral neuromodulation; Sacral neuromodulation — *afferent/forebrain mechanism, FDA indications table, PNE vs staged lead, S2/S3/S4 responses, ≥50% test, InSite, ROSETTA, PTNS technique, SUmiT, TTNS, implantable tibial, pudendal, DGNS, IVES, Brindley*

Not in the paper answer:

- Urodynamic DO does not predict success; PNE false negatives; four-contact aim; prophylaxis.
- Reprogramming ≈3 times in year 1.

1 · Evolution and mechanism

Why this matters: one device treats both overactivity and retention.

- Direct detrusor or sphincter stimulation failed (painful, local, bladder and sphincter contracted together), so stimulation moved to an **indirect route via afferents**.
- Milestones MCQ: sacral neuromodulation (SNM) ~1985 (Tanagho, Schmidt); first tibial nerve stimulation 1983 (McGuire); percutaneous tibial nerve stimulation (PTNS) from acupuncture point **SP6**; first tibial implant 2006.
- **Exclude anatomical causes first**: stones, cancer, fistula, poor compliance, stricture, prostate, prolapse.
- **Current theory**: sacral roots are mixed. Afferents reach forebrain awareness areas, which reduces pontine micturition centre (PMC) firing in overactive bladder (OAB); in underactive bladder, raised filling awareness lets the PMC switch on.
- Normally the external urethral sphincter (EUS) relaxes, then the detrusor contracts 1–2 s later. The old efferent theory is implausible: benefit needs no pelvic floor contraction, and purely sensory nerves work equally.
- **Paradox**: in retention it works mainly by **striated sphincter relaxation**, not detrusor contraction.

2 · Sacral neuromodulation

Why this matters: the implant for refractory OAB and retention.

- **Third-line** for idiopathic OAB (American Urological Association/Society of Urodynamics [AUA/SUFU] and European Association of Urology [EAU] agree).
- First get a frequency–volume diary (OAB definition ignores excess intake); exclude urinary tract infection (UTI), significant postvoid residual (PVR), stones, cancer. Urodynamic detrusor overactivity (DO) **does not predict** success.
- In **non-obstructive urinary retention (NOUR)**, PVR is the only objective outcome. Fowler syndrome predicts response; age lowers the odds.

Food and Drug Administration (FDA) approved	Not approved
Urgency urinary incontinence (UUI)	Bladder pain syndrome
Urgency–frequency	Pelvic pain syndrome
Chronic NOUR	Neurogenic bladder

- Bladder pain: offer selectively if OAB coexists. Neurogenic bladder: unapproved, but pooled success comparable. †Faecal incontinence is approved.
- Test success is ≥**50%** improvement in ≥1 diary parameter against a 3–5-day baseline; in retention, catheterisation falls ≥50%, ideally to none.
- The **S3 target** is ≈9 cm above the coccyx tip, 1–2 cm lateral. S3 gives anal "bellows" and **great-toe flexion**; S2 calf or foot rotation; S4 perineal only.
- Percutaneous nerve evaluation (PNE) is an office test with a temporary lead for 5–7 days. Its lead can migrate, giving false negatives, so conversion is lower; not used in NOUR.
- Staged: a permanent tined quadripolar lead under fluoroscopy, tested up to 4 weeks; generator in the buttock.
- Stage 2 is at 2–4 weeks; failure means lead removal. Aim for responses on all four contacts at low amplitude. Prophylaxis: cefazolin plus metronidazole, with or without MRSA cover.
- Programming: 210 µs, 10–16 Hz, at sensory threshold. ≈3 reprogrammings in year 1. Cycling equals continuous; **bilateral is not superior** (randomised controlled trial, RCT).
- Battery: non-rechargeable ≥**10 years**, rechargeable ≥15 years. Devices are whole-body **MRI**-conditional at **1.5 and 3 T**; at replacement remove all old components.
- Complications: revision ~3%, explantation ~9% at 2 years (ROSETTA). †Site or leg pain: reprogram, then revise; infection: explant.

3 · Tibial nerve stimulation

Why this matters: the least invasive third-line option.

ALGORITHM

1. Supine; insert a 34-gauge needle ≈3–4 cm above the medial malleolus, behind the tibia, at 60–90°; ground pad on the same leg.
2. Stimulate at 20 Hz, 200 µs, 0–10 mA.
3. Confirm **great-toe flexion/toe fanning** or sole tingling.
4. Give **12 weekly sessions × 30 min**, then taper to maintenance.

- It also acts by cortical reorganisation. Indications: drug-refractory or intolerant OAB; safe in neurogenic OAB (multiple sclerosis, Parkinson disease, stroke) and in children failing bladder training and drugs.
- **Transcutaneous tibial nerve stimulation (TTNS)** uses home surface electrodes; skin impedance makes it less effective than PTNS, but cheaper. Against tolterodine: equal efficacy, fewer side effects.
- Success is a ≥50% fall in urgency or UUI; **relapse is likely without maintenance**. Results are better without DO; a poor mental-health score predicts failure. PTNS response does not guarantee implant response.

4 · Other targets

Why this matters: salvage or niche options.

- **Pudendal** stimulation (S2–S4, Alcock canal) uses a posterior approach (perineal has more infection). It beat SNM in a blinded crossover and is salvage after failed SNM; not FDA approved.
- **Dorsal genital nerve**: limited by electrode site. **Sacral dermatome** (transcutaneous): raises capacity, cannot abort contractions. **Intravaginal** for stress incontinence: safe, low-quality

evidence.

- †**Intravesical stimulation** suits neurogenic underactive bladder with incomplete lesions; afferents must be intact; evidence limited.

5 · Sacral anterior root stimulation (Brindley)

Why this matters: it restores voiding in complete spinal cord injury (SCI).

- Anterior root stimulation gives voiding, defecation and erection. It is combined with **posterior sacral rhizotomy**: this abolishes reflex incontinence and autonomic dysreflexia, raises capacity and compliance, and protects kidneys.
- †Voiding is "post-stimulus": the sphincter relaxes faster than the detrusor. †Rhizotomy costs reflex erection, reflex ejaculation and residual sacral sensation.

6 · Landmark trials

Why this matters: they rank the third-line options.

Trial	Design	Result	Take-home
SUmiT	RCT, PTNS vs validated sham, 12 weeks, n = 220	Responders 55% vs 21%	First sham-proven PTNS efficacy; maintenance keeps 77% relapse-free at 3 years
InSite	RCT, SNM vs standard medical therapy, mild–moderate OAB	Success 61% vs 42% at 6 months	SNM beats continued drugs
ROSETTA	RCT, onabotulinumtoxinA 200 U vs SNM, refractory UUI women	Toxin slightly fewer leaks at 6 months; equal at 2 years; more UTI, some clean intermittent catheterisation (CIC)	Both valid third-line; registered OAB toxin dose is 100 U

★ Putting it together: voiding dysfunction decisions

A · Functional classification as treatment map

Why this matters: the failing part dictates treatment.

	Bladder	Outlet
Failure to store	Overactivity, low compliance: antimuscarinic, β3-agonist, onabotulinumtoxinA, neuromodulation, augmentation	Sphincter weakness: pelvic floor training (PFMT), duloxetine (Europe), sling, artificial urinary sphincter (AUS), bulking
Failure to empty	Underactive, acontractile: clean intermittent catheterisation (CIC); sacral neuromodulation (SNM) for non-obstructive retention	Obstruction (anatomical, dyssynergia): α-blocker, prostate/bladder-neck surgery, sphincterotomy, CIC

If neither is fixable, circumvent: CIC, indwelling catheter, diversion *(see Ch 104)*.

B · Lesion level and urodynamics

Why this matters: level predicts pattern and renal risk.

Lesion	Detrusor	Striated sphincter	Key risk
Above pons (stroke, Parkinson disease [PD], dementia)	Overactive	Synergic	Urgency incontinence; low upper-tract risk
Pons–sacral cord (spinal cord injury [SCI], multiple sclerosis [MS])	Overactive	**Dyssynergic**	High pressure damages upper tracts; autonomic dysreflexia (AD) if above T6
Sacral, cauda equina, peripheral (diabetes, pelvic surgery)	Areflexic, reduced sensation	Fixed tone	Retention, overflow; compliance may fall

KEY POINT

Dyssynergia occurs only with lesions between pons and sacral cord. A leak point or storage pressure ≥40 cmH_2O, or poor compliance, threatens the kidneys: treat *(see Ch 107, 109)*.

C · OAB ladder

Why this matters: overactive bladder (OAB) treatment is stepwise.

ALGORITHM

1. Behavioural: fluids/caffeine, bladder training, PFMT, constipation, drug review.
2. Antimuscarinic or β3-agonist; β3 or low-CNS agent in the elderly. If partial response, then combine. Check postvoid residual (PVR) in obstructed men *(see Ch 113)*.
3. If refractory, then onabotulinumtoxinA (100 U idiopathic, 200 U neurogenic; warn about CIC), tibial nerve stimulation or SNM (staged test, ≥50% improvement) *(see Ch 115)*.
4. Last resort: augmentation or diversion.

D · SCI bladder

Why this matters: drainage and surveillance protect the kidneys.

ALGORITHM

1. **Spinal shock**: catheter until reflexes return.
2. **Then** urodynamics; CIC plus antimuscarinic for high-pressure neurogenic overactivity; toxin.
3. If CIC is impossible, then sphincterotomy or diversion. Survey upper tracts lifelong.
4. **AD** (lesion above T6): hypertension, bradycardia, headache, flushing. Sit up, loosen clothing, remove the stimulus (catheter, bladder, bowel), short-acting antihypertensive *(see Ch 109)*.